

Chapter 2

Extension of EWS techniques to multivariate data

THIS IS ALMOST COMPLETE. ONLY SECTION 2.4.2 NEEDS COMPLETING. ALSO, I AM NOT SURE ABOUT THE DISCUSSION AT THE END.

Early warning signal techniques generally are applied only to univariate data, often one-dimensional time series from a single data source or model output. In an effort to study multidimensional geophysical variables we here consider cases where we have more than one time series, either because

1. there are multiple measured variables,
2. a variable is measured at multiple spacial locations,
3. both of these simultaneously.

✓ In the first case it may suffice to attempt an EWS detection on each time series individually, but one can envision a scenario where it is not possible to detect an EWS in $X(t)$ nor $Y(t)$ but, considered together in some way, an EWS is ~~visible~~ *detectable*. It may therefore be worthwhile to investigate the possibility of a two-~~or~~ more dimensional analogue to existing one-dimensional techniques. In the case of gridded data it is common to use Empirical Orthogonal Functions (EOF), also known as Principal Component Analysis (PCA), to reduce the dimensionality [von Storch and Zwiers, 2002; Jolliffe, 1986], this technique has also been more specifically applied to EWS methods [Held and Kleinen, 2004; Bathiany et al., 2013; Kwasniok, 2018]. Principal interaction and oscillation patterns (PIPs and POPs) can be considered as extensions of the EOF method and provide additional techniques for the dimension reduction of complex systems [Hasselmann, 1988]. *neither*

It has been suggested that an EWS is detectable in a dynamical system defined over a two-dimensional field, such as climatological or ecological variable modelled or measured at multiple geographical points [Guttal and Jayaprakash, 2009; Donangelo et al., 2010; Dakos et al., 2010]. There has been a large amount of work involving the use of multiple data sources spread over a geographic area, such as complex network analysis in climate [Tsonis et al., 2006; Yamasaki et al., 2008; Gozolchiani et al., 2011] which has been applied to the El Niño. In these suggested methods, the cross-covariance or cross-correlation is calculated between points on a grid, rather than the temporal autocorrelation (or related properties) being calculated. The assumption here is that the correlation between neighbouring points will increase close to a tipping point.

Ludescher et al. [2013, 2014] calculate the cross-covariance of a variable, at a specific time slice, between points inside a region of interest (in this case, the El Niño basin) and points outside. In contrast, simply using the EOF method to reduce the dimension of such a system will not give specific spatial information such as the location of the tipping point or the direction of its progression in the case that a tipping at one location precipitates a tipping at other locations. Bathiany et al. [2013] present a method to detect "hotspots", locations where a tipping point is likely to occur, or rather locations where critical slowing down, and therefore loss of stability is most apparent. The method involves partitioning the locations ("elements") and, within each partition, discovering which elements contribute the most to the autocorrelation. This is done by combining the elements in different permutations, projecting onto the EOFs, and calculating the lag-1 autocorrelation of the resulting signal. Elements in each part are removed according to system-specific selection criteria and the remaining elements are re-partitioned. The process is repeated until only one partition remains, which is the slowing down hotspot where susceptibility to perturbations is large. The method is applied successfully to a highly idealised model of a vegetation cover.

In this chapter we review and analyse techniques which may be used to calculate an EWS with an input of two variables, namely the methods presented by Williamson and Lenton [2015] (in section 2.1), and the use of EOFs for dimension-reduction (in section 2.2). These two techniques are applied, in section 2.3, to multivariable time series data from models of three simple dynamical systems (two are examples taken from Williamson and Lenton [2015], and the results are successfully reproduced). The third example is the Van der Pol oscillator, which allows us to test these methods in a novel system. In chapter 3 these same methods will be applied to real measurements of physical variables, alongside the familiar one-dimensional techniques (see chapter 1).

In section 2.4 the method of using EOF for dimension reduction prior to the application of an EWS indicator is tested analytically, thus providing a step towards a mathematically

rigorous understanding of the commonly used method. In section 2.5 we present an alternative formulation of the EOF method for dimension reduction.

In section 2.6 we present a method to visualise early warning signals in a dynamical system over a 2D field, and apply this method to data from a simple model of a progressing "front", in which the variable is modelled at discrete grid points. Again, in chapter 3, this method will be applied to real measured physical variables, that is, the time series of variables measured at geographically separated locations in the vicinity of an approaching tropical cyclone,

monitoring of atmospheric pressure with

2.1 Multivariate extension of the autocorrelation function

Williamson and Lenton [2015] introduce a method to anticipate bifurcations in dynamical systems with more than one variable. The method is ~~intentioned~~ as a higher-dimensional analogue of the one-dimensional ACF1 indicator. Indeed, we note the similarities between the equation used to calculate the lag-1 autocorrelation coefficient a in the one dimensional time series $\{x_t\}$:

$$a = \frac{\sum_t (x_{t+1}x_t) - \bar{x}^2}{\sum_t (x_t^2) - \bar{x}^2}, \quad (2.1)$$

and the equation given by the authors:

$$A = \left[\sum_t (\mathbf{x}_{t+1}\mathbf{x}_t^\top) - \bar{\mathbf{x}}^2 \right] \left[\sum_t (\mathbf{x}_t\mathbf{x}_t^\top) - \bar{\mathbf{x}}^2 \right]^{-1}, \quad (2.2)$$

where $\{\mathbf{x}_t\}$ is a multivariate time series. By linearising and discretizing, one may approximate the one-dimensional time series $\dot{x} = f(x) + \varepsilon(t)$ as the autoregressive system

$$x_t = ax_{t-1} + c + \varepsilon_t, \quad (2.3)$$

where c is a scalar value and ε_t is Gaussian noise. An AR(1) model of a system may therefore be constructed based only on time series data, by calculating the parameter a using equation 2.1, that is, the lag-1 autocorrelation coefficient. In the same way, the matrix A may be used to model an N -dimensional system $\dot{\mathbf{x}}^{(i)} = f(x^{(1)}, \dots, x^{(N)}) + \varepsilon^{(i)}$ using the analogous form

$$\mathbf{x}_t = A\mathbf{x}_{t-1} + \mathbf{c} + \varepsilon_t, \quad (2.4)$$

where $\mathbf{c}$ is a constant vector and ε_t is a vector with each element being an independent Gaussian white noise term. In this linear system a tipping point occurs when an eigenvalue of A has the value one. It is therefore changes in the eigenvalues of A that must be studied and these are

✓ calculated easily once the ~~the~~ matrix A has been found according to equation 2.2, from time series data. However, it is more useful to study the eigenvalues of the Jacobian of the system equations, since their behaviour is already well studied in many contexts. For example, a Hopf bifurcation is characterised by a conjugate pair of Jacobian eigenvalues crossing the imaginary axis [Strogatz, 2014]. Tracking the changes in the imaginary part of one of these eigenvalues as it approaches zero therefore gives an early warning signal of the Hopf bifurcation.

Given a system of equations, for example the two-dimensional system

$$\begin{aligned} \dot{x} &= f(t, x, y) \\ \dot{y} &= g(t, x, y) \end{aligned} \quad (2.5)$$

✓ which has a stable centre at the point $\mathbf{x}_* = (x_*, y_*)$, we can calculate the Jacobian matrix explicitly, evaluated at ~~the~~ this point:

$$J(\mathbf{x}_*) = \begin{pmatrix} \left. \frac{df}{dx} \right|_{\mathbf{x}_*} & \left. \frac{df}{dy} \right|_{\mathbf{x}_*} \\ \left. \frac{dg}{dx} \right|_{\mathbf{x}_*} & \left. \frac{dg}{dy} \right|_{\mathbf{x}_*} \end{pmatrix}. \quad (2.6)$$

This allows us to study the effect of a small perturbation in x or y and, as noted, the eigenvalues of the Jacobian matrix are well studied in many dynamical systems [Thompson et al., 2002].

✓ However, we are interested in detecting and anticipating tipping points in time series, not analytically defined systems. Williamson and Lenton [2015] provide a method for estimating the eigenvalues λ_k of the Jacobian, given a time series. Rather than attempting to calculate the λ_k directly from a time series, the authors calculate the matrix A directly using equation 2.2 and then refer to the discretised system (equation 2.4), in which $A = \mathbf{I} + J(\mathbf{x}_*)\Delta t$. The first order approximation

$$A = \mathbf{I} + J(\mathbf{x}_*)\Delta t \approx \exp(J(\mathbf{x}_*)\Delta t) \quad (2.7)$$

is used to recover the Jacobian eigenvalues λ_k from the eigenvalues $a_k = |a_k| \exp(i\phi)$ of A , with the relations

$$\begin{aligned} \Re(\lambda_k) &= \frac{1}{\Delta t} \ln |a_k|, \\ \Im(\lambda_k) &= \frac{1}{\Delta t} \phi_k. \end{aligned} \quad (2.8)$$

you may like to replace ϕ with ϕ_{arphi}

The eigenvalues λ_k are then, themselves, tipping point indicators. To obtain an EWS, the method then involves calculating the matrix A from the given time series data in a sliding

✓ window, similarly to the one-dimensional ACF indicator ~~(etc.)~~, and recovering the Jacobian eigenvalues using equation 2.8.

We note that this method is only effective as an EWS if the type of expected tipping, particularly a bifurcation, is known in advance, since it is necessary to know which eigenvalue to study ~~and how to~~ use that eigenvalue as an indicator. However, an obvious increase or decrease in either eigenvalue may provide an indication that ~~something~~ is happening, and could therefore lend confidence to an EWS derived by another method. The method, however, would certainly be useful in the detection and classification of tipping points, when applied retrospectively. In systems of more than two variables the situation is more acute because of the increased number of eigenvalues. Identifying conjugate pairs may lead to some insight as to which could be indicators (depending of the type of bifurcation), and will reduce the number of distinct eigenvalues.

Analysis of a Hopf bifurcation

✓ In order to analyse this method further, we take, as an example, the Van der Pol oscillator

$$\begin{aligned}\dot{x} &= \mu \left(x - \frac{1}{3}x^3\right) + y, \\ \dot{y} &= -x,\end{aligned}\tag{2.9}$$

which experiences a Hopf bifurcation at $\mu = 0$ if μ ~~is made to vary~~ ^{varies}. That is, the eigenvalues are a complex conjugate pair whose imaginary parts cross the imaginary axis at $\mu = 0$ [Strogatz, 2014]. We shall integrate this system, with added noise, numerically in section 2.3 and apply the EWS method described here to the resulting time series. Here we consider only the analytic process of the method.

We see that the system has a stable point at $\mathbf{x}_* = (0, 0)$ and the Jacobian is found to be

$$J(\mathbf{x}_*) = \begin{pmatrix} \mu(1-x^2) & 1 \\ -1 & 0 \end{pmatrix} \bigg|_{(0,0)} = \begin{pmatrix} \mu & 1 \\ -1 & 0 \end{pmatrix}\tag{2.10}$$

with eigenvalues

$$\lambda = \frac{1}{2} \left(\mu \pm \sqrt{\mu^2 - 4} \right)\tag{2.11}$$

$$= \frac{\mu}{2} \pm i \frac{\sqrt{4 - \mu^2}}{2} \quad \text{where } |\mu| < 2.\tag{2.12}$$

ignore this

We obtain the matrix A by discretising according to the Euler method:

$$\begin{pmatrix} x_{t+1} \\ y_{t+1} \end{pmatrix} = \begin{pmatrix} x_t \\ y_t \end{pmatrix} + \Delta t \begin{pmatrix} \mu(x_t - x_t^3/3) + y_t \\ -x_t \end{pmatrix}. \quad (2.13)$$

We use the linearisation $x^3 = 0$ to continue,

$$\begin{pmatrix} x_{t+1} \\ y_{t+1} \end{pmatrix} = \begin{pmatrix} x_t \\ y_t \end{pmatrix} + \Delta t \begin{pmatrix} \mu x_t + y_t \\ -x_t \end{pmatrix} \quad (2.14)$$

$$= \begin{pmatrix} 1 + \Delta t \mu & \Delta t \\ -\Delta t & 1 \end{pmatrix} \begin{pmatrix} x_t \\ y_t \end{pmatrix}. \quad (2.15)$$

The matrix in equation 2.15 is the matrix A , with eigenvalues

$$a = \frac{2 + \Delta t \mu}{2} \pm i \frac{\Delta t \sqrt{4 - \mu^2}}{2}, \quad \text{where } |\mu| < 2 \quad (2.16)$$

or $a = |a| \exp(i\phi)$ where

$$\begin{aligned} |a| &= \sqrt{\Delta t^2 + \Delta t \mu + 1}, \\ \phi &= \arctan\left(\frac{\Delta t \sqrt{4 - \mu^2}}{2 + \Delta t \mu}\right). \end{aligned} \quad (2.17)$$

These eigenvalues of A can now be used to reconstruct the eigenvalues of the Jacobian. Referring to equations 2.8, and using the power series expansions $\arctan(x) = x - x^3/3 + \dots$ and $\ln(x + 1) = x - x^2/2 + \dots$, we find

$$\mathbb{R}(\lambda) = \frac{1}{\Delta t} \ln |a| = \frac{\mu}{2} + \frac{\Delta t}{2} + \Delta t \frac{\mu^2}{4} + \mathcal{O}(\Delta t^2), \quad (2.18)$$

$$\mathbb{I}(\lambda) = \frac{1}{\Delta t} \phi = \frac{\sqrt{4 - \mu^2}}{2 + \Delta t \mu} + \mathcal{O}(\Delta t^2). \quad (2.19)$$

If we use the approximation $\Delta t = 0$, which is valid for very small Δt , these agree exactly with the Jacobian eigenvalues found in equation 2.12. Thus, from the linearised system $\mathbf{x}_{t+1} = \mathbf{A}\mathbf{x}_t$ in equation 2.15, we have reconstructed the Jacobian eigenvalues, evaluated at the stable centre, of the non-linearised system (equation 2.9), with only the assumption that Δt is close to zero.

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2.2 Dimension reduction using EOFs

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In this section we study the technique of dimension reduction using empirical orthogonal functions (EOFs), also known as principal components analysis (PCA). This technique is widely used in geosciences and meteorology to aid the study of a large, multivariate data set [Bjornsson and Venegas, 1997; Stephenson and Benestad, 2000; Shaw, 2009]. [Held and Kleinen [2004] use the EOF technique prior to applying the ACF(1) indicator to a set of gridded time series, relying on the observation made in [Scheffer et al., 2009] that an increase in autocorrelation is typically accompanied by an increase in variance. [Held and Kleinen [2004] use this observation to argue that in a multi-dimensional system one should project onto the first EOF (the basis vector for which the variance of the system is maximised) as this is the one-dimensional basis in which the rise in autocorrelation (and also variance) will occur.

paper of 2004
cannot rely on

paper of 2009

We note that the basis in which the variance is maximal may not be the same as the basis in which the variance is increasing. At least, this is not obvious a priori. In some dynamical systems it may be that while autocorrelation rises before a tipping point, variance does not [Gsell et al., 2016]. This observation is investigated further in section 2.4.

In this section we explain the EOF method for dimension reduction. In section ?? we apply this method to three two-variable time series prior to using the one-dimensional ACF1, DFA and PS indicators.

lit

Empirical orthogonal functions

The EOF method involves projecting a data matrix Y of N time series onto a different basis to obtain N new time series. This basis is orthogonal and is constructed so that projecting Y onto the first basis vector maximises the variance of the projection. The second basis vector is such that projecting Y onto it will maximise the variance given that it must be orthogonal to the first, and so on. Only the first one or two EOF scores may capture most of the variance of the whole data set of hundreds of time series.

We consider N observations $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N$ of a P -dimensional system, collected in the $N \times P$ matrix X , shown in equation 2.20.

$$X = \begin{bmatrix} \mathbf{x}_1^T \\ \vdots \\ \mathbf{x}_N^T \end{bmatrix} = \begin{bmatrix} x_{11} & \dots & x_{1P} \\ \vdots & \ddots & \vdots \\ x_{N1} & \dots & x_{NP} \end{bmatrix}. \quad (2.20)$$

also
[Lima et al 2012]

✓

The method requires a mean-centred series, it is common to replace the series $[\mathbf{x}_n]_{n=1}^N$ with a new set of series $[\mathbf{y}_n]_{n=1}^N$, where the i^{th} element of $\mathbf{y}_n$, which we call $\mathbf{y}_n^{(i)}$, is given by

$$\mathbf{y}_n^{(i)} = \mathbf{x}_n^{(i)} - \frac{1}{N} \sum_{j=1}^N \mathbf{x}_j^{(i)}, \quad (2.21)$$

so that a new matrix Y is created from the N time series $\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_N$, similar to the matrix X in equation 2.20. We then project Y onto a different basis to obtain the matrix T of EOF scores. The first column of T being the first EOF score. We now calculate the matrix W of basis vectors, these are the eigenvectors of the covariance matrix

$$C = \frac{1}{N-1} Y^T Y. \quad (2.22)$$

The eigenvector corresponding to the largest eigenvalue is the first column of the matrix W , and so on. Then we are able to calculate $T = YW$. Often, only the first one or two EOF scores are required, therefore we may create a smaller matrix W whose columns are the first one or two eigenvectors of C (those corresponding to the ~~one or two~~ largest eigenvalues).

The method described above may be used to reduce the dimension of a system from many hundreds or thousands, to a ten or fewer, particularly in cases when the multidimensionality arises from a variable having been measured or modelled at discrete points of a grid. In this study, we are particularly interested in reducing the dimension to just one, so that the one-dimensional EWS indicators described in chapter 1 may be applied. The choice of W is therefore simple, the column vector which is the eigenvector of C corresponding to the largest eigenvalue. Having obtained the one-dimensional series $T = YW$, in this case in the form of an $N \times 1$ matrix, the sliding-window or "fingerprinting" indicator methods may be applied with an appropriate window size.

2.3 Application of methods to bifurcating dynamical systems

In this section we reproduce the results presented by Williamson and Lenton [2015], in which the method described in section 2.1 is applied to two two-dimensional dynamical systems: one experiencing a homoclinic bifurcation and one experiencing a Hopf bifurcation. We also apply the method to a Van der Pol oscillator which experiences a Hopf bifurcation, as an independent test of the method. Since we will be refer to these systems often, we will call them systems A, B and C respectively for brevity, wherever the meaning is obvious.

In addition, we provide a comparison between this method and the method of using EOFs for dimension reduction Held and Kleinen [2004]. We reduce the dimension of the time series

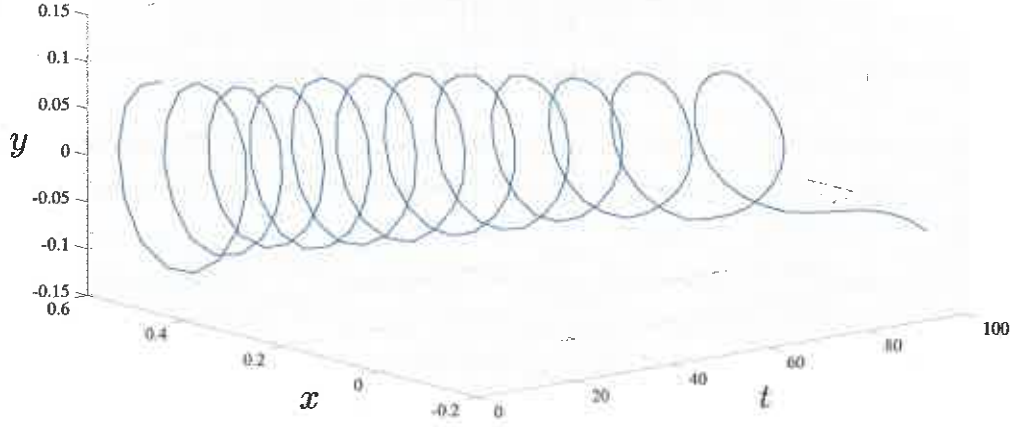


Fig. 2.1 Showing a single realisation of the system in equation 2.23 with initial point $(\sqrt{0.2}, 0.1)$. The system spirals around the moving centre at $(\sqrt{\alpha}, 0)$, α is given in equation 2.25, until $t = 0$, $\alpha = 0$, when the bifurcation occurs and the system tends towards infinity in the negative x and negative y directions.

data obtained from the dynamical systems from two to one, using the EOF method, in order to apply the one-dimensional ACF1, DFA and PS indicators described in chapter 1.

The dynamical systems presented in this section as systems of ODEs are integrated numerically using MatLab's `ode45` solver [Shampine and Reichelt, 1997] which is based on the Dormand-Prince explicit Runge-Kutta (4,5) method [Dormand and Prince, 1980]. This is used consistently throughout.

System experiencing a Homoclinic bifurcation (System A)

We consider the system given by equations

$$\begin{aligned}\dot{x} &= y + \varepsilon^{(x)}, \\ \dot{y} &= \alpha - x^2 + \varepsilon^{(y)},\end{aligned}\tag{2.23}$$

where $\varepsilon^{(x)}$, $\varepsilon^{(y)}$ are white noise terms. If we ignore these noise terms, we find that the system has steady state solutions at $(x, y) = (\pm\sqrt{\alpha}, 0)$. The Jacobian of the system is given by

$$J(x, y) = \begin{pmatrix} 0 & 1 \\ -2x & 0 \end{pmatrix}\tag{2.24}$$

and has eigenvalues $\lambda = \pm\sqrt{-2x}$. The eigenvalues at the stable solutions are

✓ at $(-\sqrt{\alpha}, 0)$: $\lambda = \pm(4\alpha)^{1/4}$, indicating that this is a saddle,

✓ at $(+\sqrt{\alpha}, 0)$: $\lambda = \pm i(4\alpha)^{1/4}$, indicating that this is a stable centre.

Williamson and Lenton [2015] allow α to vary and to approach zero from above. At $\alpha = 0$ the two stable points collide resulting in a homoclinic bifurcation and there are no stable solutions for $\alpha < 0$.

The authors consider a point at or close to the stable centre $(+\sqrt{\alpha}, 0)$, with noise terms $\varepsilon^{(x)}, \varepsilon^{(y)}$ in equation 2.23 having standard deviation of 0.01. At this centre the eigenvalues of the Jacobian have zero real part but imaginary part of $\pm(4\alpha)^{1/4}$. If we take the eigenvalue with positive imaginary part we expect to see it decrease as α decreases to zero. This is the early warning signal of the bifurcation.

We integrate the system in equations 2.23 from $t = 0$ to $t = 100$, using a varying parameter

$$\alpha(t) = 0.2 - 0.002t, \quad (2.25)$$

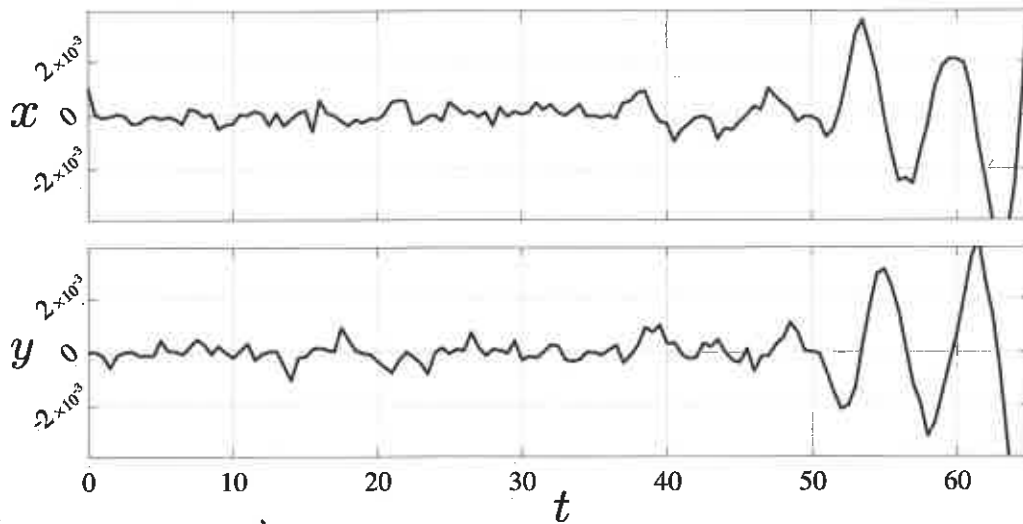
so that the bifurcation when $\alpha = 0$ will occur at $t = 100$. Noise terms $\varepsilon^{(x)}, \varepsilon^{(y)}$ are independent with zero mean and standard deviation 0.01 and the solution is sampled at intervals of $\Delta t = 0.5$. The result of one such integration is plotted in figure 2.1.

We now attempt to provide an early warning signal of the impending bifurcation. The autocorrelation matrix A is calculated using equation 2.4 in a sliding window of 100 points. The Jacobian eigenvalues are recovered using equations 2.8 and the real and imaginary parts of the first eigenvalue (the eigenvalue with positive imaginary part) are calculated. This procedure is applied to ten realisations of the system, and the mean is shown in figure 2.4a,b with error bars of one standard deviation. The figure shows a successful reproduction of the results of Williamson and Lenton [2015]: the real part of the eigenvalue is ≈ 0 with little variability, as expected for $x > 0$ and the decreasing imaginary part of the eigenvalue predicates the bifurcation, as predicted by the analysis, as $\alpha \rightarrow 0$.

As a comparison, the ACF1 and DFA indicators are applied to the same data using a 100 point window size and after reducing the dimension of the system using EOFs. The results are shown in figure 2.4c,d. Both indicators show a general upwards trend prior to the bifurcation at $t = 100$, but the ACF1 indicator contains more variability.

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✓ Fig. 2.2 Showing a single realisation of the system in equation 2.26 with initial point $(0,0)$. The system oscillates about the stable point $(0,0)$ until $t = 58$ when the bifurcation occurs and the stability is lost.

System experiencing a Hopf bifurcation (System B)

✓ We now consider the second example given by Williamson and Lenton [2015], the system given by the polar equations

$$\begin{aligned}\dot{r} &= \mu r - r^3 \\ \dot{\theta} &= 1 + r^2.\end{aligned}\quad (2.26)$$

The system is a classic example of a supercritical Hopf bifurcation. The normal form of the Hopf bifurcation is

$$\dot{z} = z((\lambda + i) + b|z|^2), \quad (2.27)$$

✓ where z and $b = \alpha + i\beta$ are complex numbers. If we rewrite the system in equation 2.26 according to $z = \exp(i\theta)$, we get

$$\dot{z} = \dot{r}e^{i\theta} + ir\dot{\theta}e^{i\theta} \quad (2.28)$$

$$= re^{i\theta} \left(\frac{\dot{r}}{r} + i\dot{\theta} \right) \quad (2.29)$$

$$= re^{i\theta} ((\mu - r^2) + i(1 + r^2)) \quad (2.30)$$

$$= z((\mu + i) + (i - 1)|z|^2). \quad (2.31)$$

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✓ The system has stable solution at $r = 0$ for $\mu < 0$. At $\mu = 0$ a bifurcation occurs and $r = 0$ becomes unstable, whilst two stable solutions for r are created at $r = \pm\sqrt{\mu}$. Since θ increases

at a constant rate $\dot{\theta} = 1 + \mu$, or almost constant if an additional noise term is introduced into equation 2.26, this results in a stable limit cycle with radius $r = \sqrt{\mu}$.

We may also investigate the nature of the bifurcation by studying the eigenvalues of the Jacobian of the system equations. In Cartesian rather than polar coordinates we have

$$J(x, y) = \begin{pmatrix} \mu - (x+y)^2 - 2x^2 & -1 - (x+y)^2 - 2y^2 \\ 1 + (x-y)^2 + 2x^2 & \mu - (x-y)^2 - 2y^2 \end{pmatrix}. \quad (2.32)$$

At the stable point $r = 0$ (i.e. $(x, y) = (0, 0)$), the matrix $J(x, y)$ has complex conjugate eigenvalues $\lambda = \mu \pm i$. The system therefore experiences a Hopf bifurcation as μ approaches zero from below, characterised by the Jacobian eigenvalues moving from the negative-real to the positive-real half of the complex plane.

We reproduce the numerical results of Williamson and Lenton [2015] using a varying parameter

$$\mu(t) = 0.05t - 2.9 \quad (2.33)$$

and integrate the system from $t = 0$ to $t = 60$ with the solution given at intervals $\Delta t = 0.2$. The bifurcation at $\mu = 0$ will occur at $t = 58$. Gaussian noise terms $\epsilon^{(r)}$ and $\epsilon^{(\theta)}$ are also added to the system equations for r and θ respectively (equation 2.26), both with zero mean and standard deviation 0.01. The result is shown in figure 2.2.

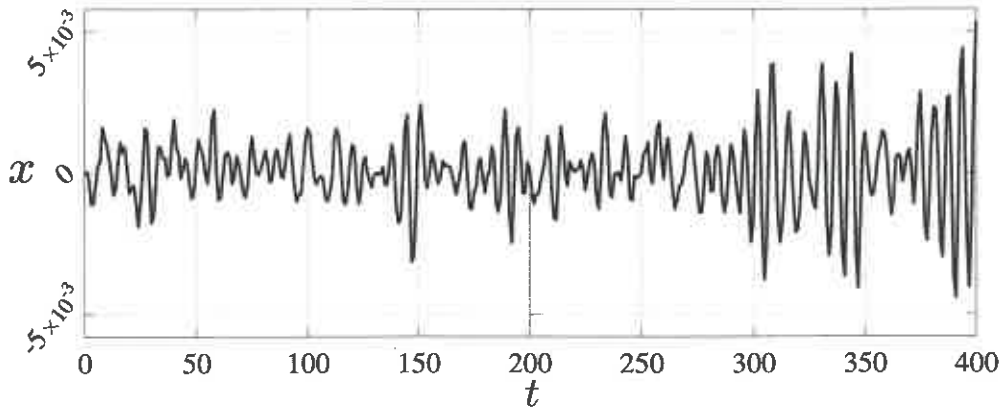
As in the previous example, the autocorrelation matrix A is calculated using equation 2.4 in a sliding window of 100 points. The Jacobian eigenvalues are recovered using equation 2.8 and the real and imaginary parts of the first eigenvalue are plotted in figure 2.5a,b.

In this example, as in the previous, we are able to successfully reproduce the results of Williamson and Lenton [2015].

The Van der Pol oscillator (System C)

Here, we consider another example of a Hopf bifurcation, allowing us to apply the method introduced in [Williamson and Lenton, 2015] to a novel system. We consider the Van der Pol oscillator [Van der Pol, 1926] given by

$$\ddot{x} - \mu(x - x^2)\dot{x} + x = \epsilon, \quad (2.34)$$



✓ Fig. 2.3 Showing a single realisation of the Van der Pol system in equation 2.35 with initial point $(0,0)$. The system oscillates about centre at $(0,0)$ until $t = 380$, $\mu = 0$, when the Hopf bifurcation occurs and stability is lost.

where the stochastic forcing term ε is white noise with standard deviation 0.01. We may also write equation 2.34 as a coupled system of first order ODEs:

✓

$$\begin{aligned}\dot{x} &= \mu \left(x - \frac{1}{3}x^3 \right) + y, \\ \dot{y} &= -x + \varepsilon.\end{aligned}\tag{2.35}$$

The system has a stable equilibrium point ($\dot{x} = \dot{y} = 0$) at $(x, y) = (0, 0)$ for $\mu < 0$ which becomes the centre of a stable limit cycle for $\mu > 0$. The Jacobian of this system is calculated:

$$J(x, y) = \begin{pmatrix} \mu(1 - x^2) & 1 \\ -1 & 0 \end{pmatrix}.\tag{2.36}$$

Evaluated at the stable point $(0, 0)$, the two complex-conjugate eigenvalues of the Jacobian are $\lambda = \frac{1}{2}(\mu \pm \sqrt{\mu^2 - 4})$. We therefore expect that the real part of both of the eigenvalues, equal to $\mu/2$, will approach zero as μ approaches zero from below, i.e. as the system approaches the bifurcation.

We integrate the system in equations 2.35 from $t = 0$ to $t = 400$ using a time dependent parameter

$$\mu(t) = -0.38 + 0.001t,\tag{2.37}$$

✓ so that the bifurcation $\mu = 0$ will occur at $t = 380$. Using the method described, we estimate the first eigenvalue of the Jacobian in a sliding window of 100 points. The mean (with error bars of one standard deviation) of 10 such experiments is shown in figure 2.6. We note that the

real part of the eigenvalue increases as expected and provides an early warning signal of the impending Hopf bifurcation.

2.3.1 Application of the novel PS indicator

We also calculate the PS indicator in the three systems described in this section, in addition to the ACF1 and DFA indicators and the Jacobian eigenvalues as shown in figures 2.4-2.6. Systems A, B and C (see equations 2.23, 2.26 and 2.35) are integrated and the PS indicator is calculated, in all three cases, using a sliding window of 100 points, similarly to the other indicator methods. This is done ten times for each system and the mean of the PS indicator is found, as shown in figure 2.7. We see that the PS indicator does not provide an EWS for the bifurcation in systems A and C, and the signal is very weak, obscured by its variability, in system B. In all three cases the DFA indicator, as presented in figures 2.4-2.6, provides a clearer EWS.

2.4 Justification of EOF dimension reduction

In this section we attempt to justify the use of EOF for dimension reduction in an early warning signal context, as presented in section 2.2. By taking the first EOF score of a set of time series, one is projecting on the basis such that the variance is maximised. However, one may favour a dimension which happens to be noisy, having a large variance but no interesting features indicative of a tipping point, whereas viewing the data set along another dimension, which contains lower variability, may yield a useful early warning signal. It is clear that including a single time series composed of white noise with very large variance will make the first EOF score useless; it should be hoped that such cases can be removed before applying the EOF method. However, even in less artificial cases, it is not obvious that viewing a system along a dimension such that the variance is maximised will provide the best EWS in a tipping points context. ~~If a good EWS is needed,~~ one would rather look at the increase in variance, or autocorrelation, rather than the variance itself.

The essential hypothesis relied upon when using EOF for dimension reduction prior to detecting an EWS, is that the basis in which the variance is maximised (using the EOF method) is similar to the basis in which the increase in the variance is maximised. To investigate this hypothesis we consider the P -dimensional discrete-time dynamical system $\mathbf{x}_n$ described by

$$\mathbf{x}_n = B\mathbf{x}_{n-1} + S\eta_n, \quad (2.38)$$

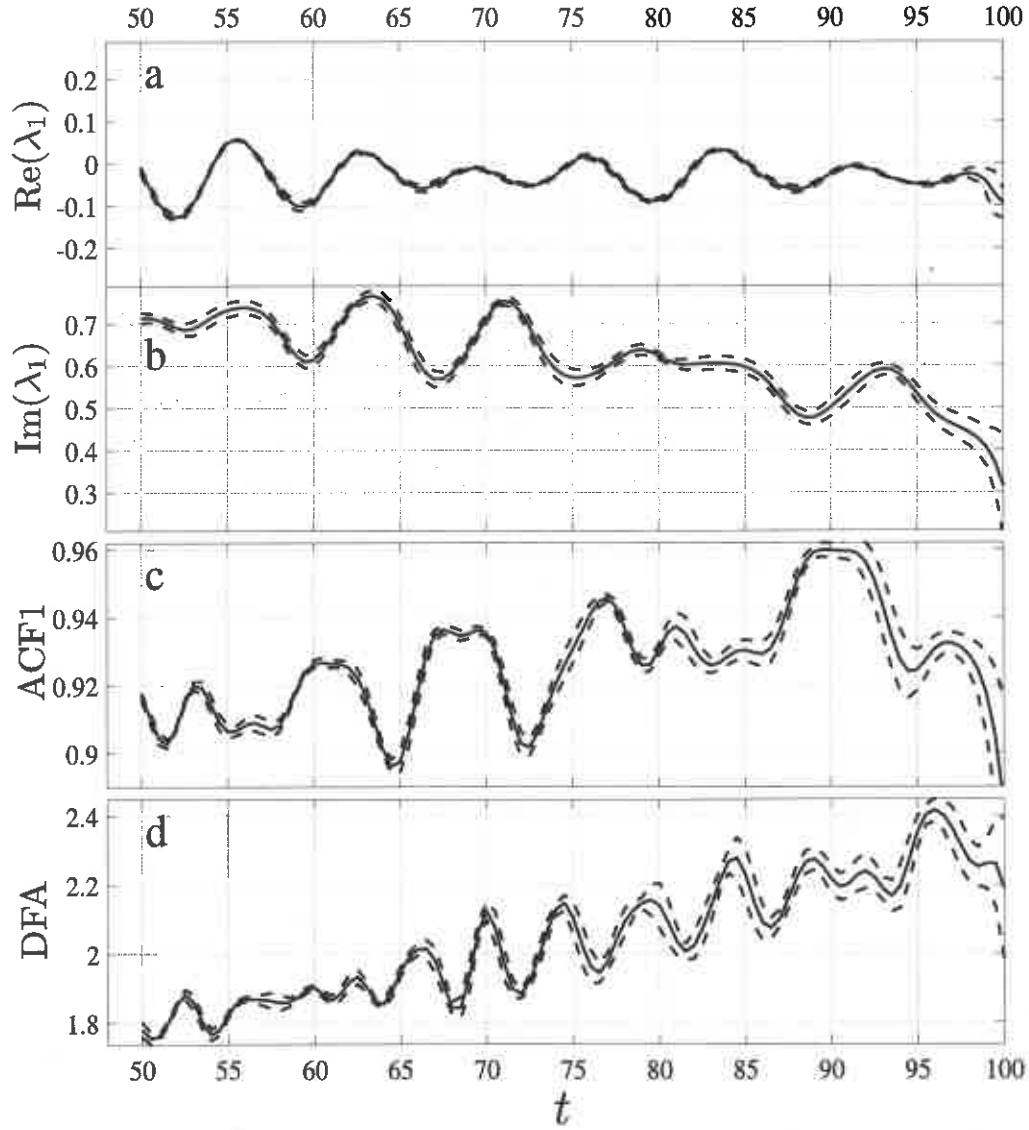
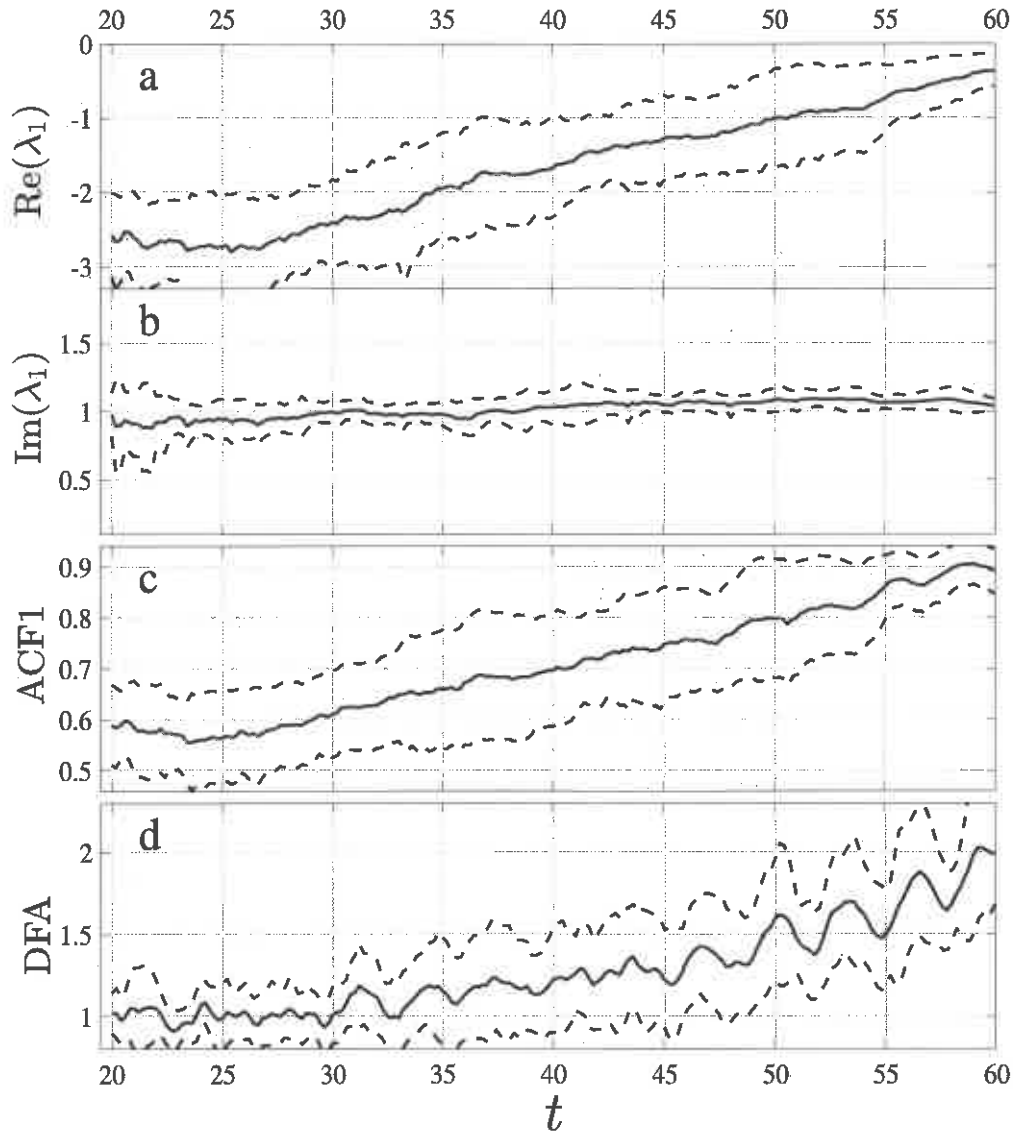


Fig. 2.4 Showing different EWS indicators applied to System A (equation 2.23) which experiences a homoclinic bifurcation at $t = 100$. Panels **a** and **b** show the real and imaginary parts of the first reconstructed eigenvalue of the Jacobian matrix, as in equation 2.8. Panels **c** and **d** show the ACF1 and DFA indicators calculated with a window size of 100 points, applied to the one-dimensional time series obtained by applying EOF dimension reduction to the data. The system was integrated ten times and the mean over ten data sets is plotted, along with error of one standard deviation (dashed lines).



✓ Fig. 2.5 Showing different EWS indicators applied to System B (equation 2.26) which experiences a Hopf bifurcation at $t = 58$. Panels **a** and **b** show the real and imaginary parts of the first reconstructed eigenvalue of the Jacobian matrix, as in equation 2.8. Panels **c** and **d** show the ACF1 and DFA indicators calculated with a window size of 100 points, applied to the one-dimensional time series obtained by applying EOF dimension reduction to the data. The system was integrated ten times and the mean over ten data sets is plotted, along with error of one standard deviation (dashed lines).

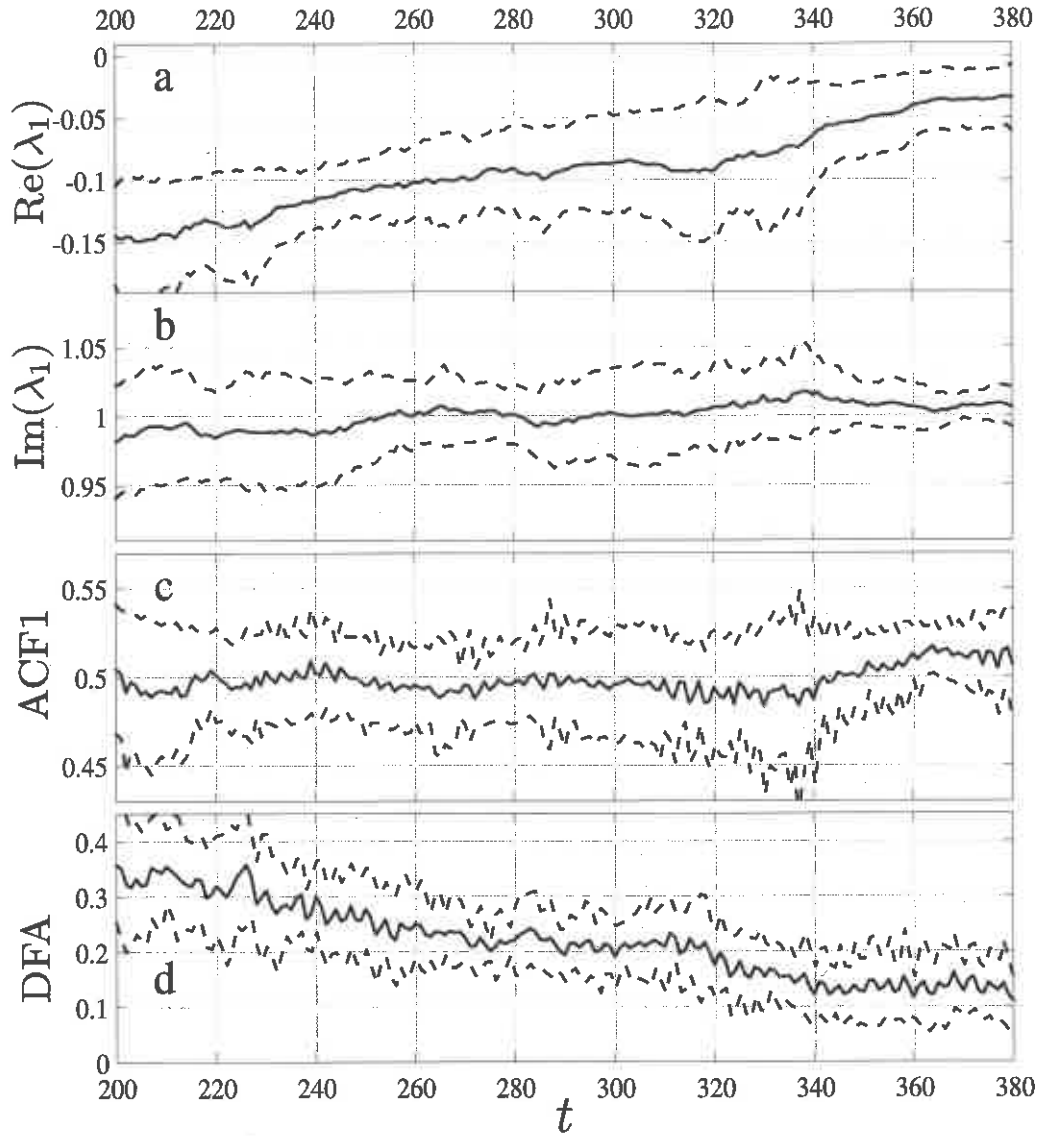


Fig. 2.6 Showing different EWS indicators applied to System C, the Van de Pol oscillator (equation 2.35) which experiences a Hopf bifurcation at $t = 380$. Panels **a** and **b** show the real and imaginary parts of the first reconstructed eigenvalue of the Jacobian matrix, as in equation 2.8. Panels **c** and **d** show the ACF1 and DFA indicators calculated with a window size of 100 points, applied to the one-dimensional time series obtained by applying EOF dimension reduction to the data. The system was integrated ten times and the mean over ten data sets is plotted, along with error of one standard deviation (dashed lines).

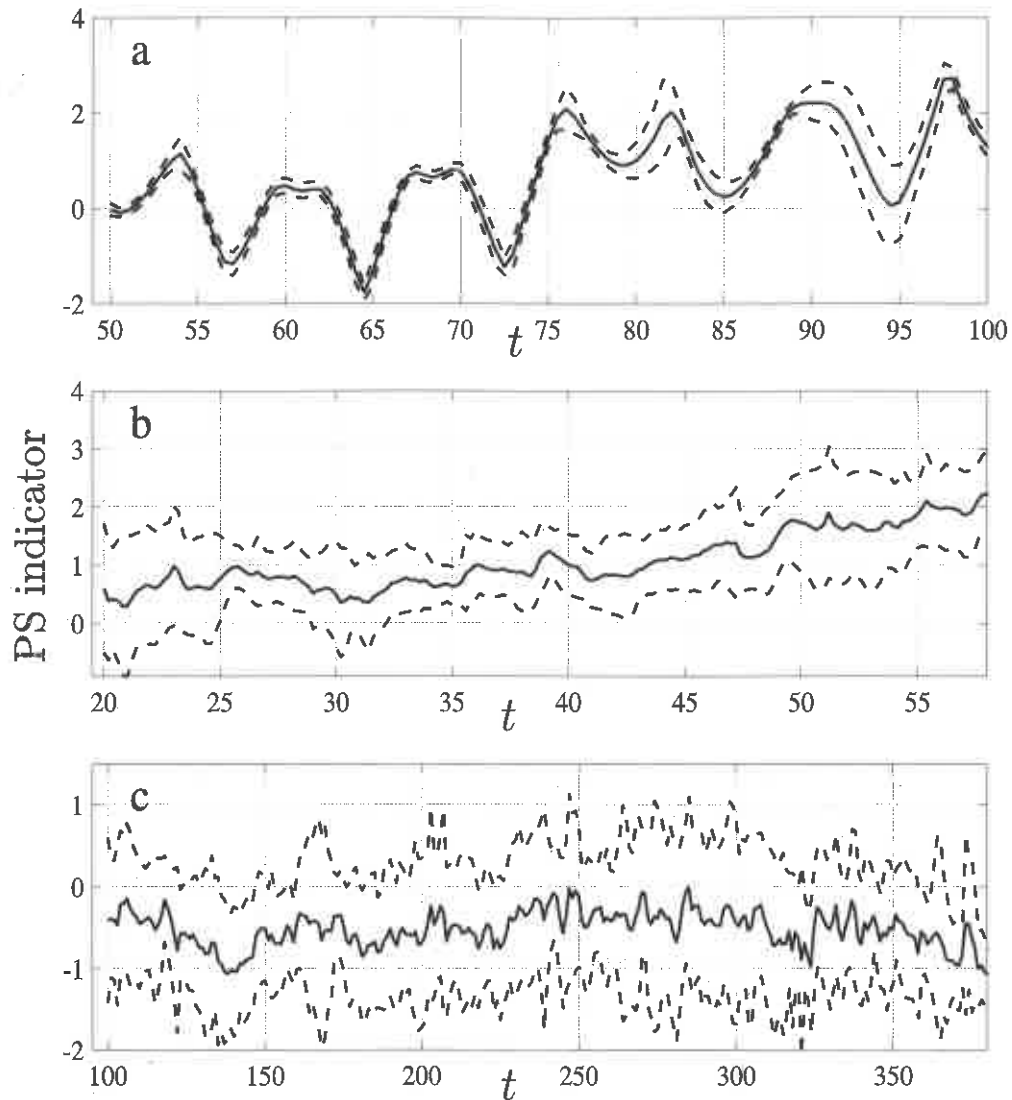


Fig. 2.7 Showing the PS indicators applied to the three systems described in this section after reducing each system to one dimension using the EOF method. The results for systems A, B and C are shown in panels a, b and c respectively. In each plot the bifurcation occurs at the last point on the t -axis: $t = 100$, $t = 58$ and $t = 380$ respectively. Each system was integrated ten times and the mean over ten data sets is plotted, along with error of one standard deviation (dashed lines).

✓ where B, S are symmetric $P \times P$ matrices and the η_n are column vectors with each element ^{being} independent Gaussian white noise. If $\mathbf{x}$ is one-dimensional ($P = 1$), then the system is an AR(1) model. If $P > 1$, one may want to study only the first EOF score in the hope that this captures most of the interesting behaviour of the system [Held and Kleinen, 2004].

In the simple case where S is the zero matrix, we observe that, as n goes to infinity, for each eigenvalue λ_k of B such that $|\lambda_k| < 1$, the system goes to zero in the direction of the corresponding eigenvector. Similarly, for each eigenvalue λ_k of B such that $|\lambda_k| > 1$, the system goes to infinity in the direction of the corresponding eigenvector.

✓ In this section we consider contracting systems, where all eigenvalues have magnitude less than one. In this case, we have a system which spirals towards zero, and approaches zero fastest in the direction of the eigenvector corresponding to the smallest eigenvalue. We then consider a matrix B constructed such that one of its eigenvalues λ_1 increases over time, so that there is a tipping point at $\lambda_1 = 1$ when the system diverges to infinity in the direction of the eigenvector corresponding to λ_1 . It is in the direction of this eigenvector that the tipping point will be most visible and, therefore, projecting the system onto this eigenvector will give the one-dimensional system in which detecting the tipping point will be easiest. If dimension reduction using the EOF method is to be used prior to using one-dimensional EWS methods, one ^{would} expect the vector onto which we project the system (the first column of W , see section ??) to be in the same direction as the eigenvector of B corresponding to the eigenvalue which is approaching the value one.

✓ This becomes more obvious if we say that B is diagonal, which is equivalent to saying that B is diagonalisable, and $S = 0$. Say $B = \text{diag}(b_1, b_2, \dots, b_n)$, then $b_1, b_2, \dots, b_n$ are the eigenvalues of B and the system becomes P separate one-dimensional systems,

$$\mathbf{x}_n^{(i)} = b_i^n \mathbf{x}_{n-1}^{(i)} \quad i = 1, 2, \dots, P. \quad (2.39)$$

for bigger space use \quad command

If $b_m = \min_i |b_i|$ the system will go to zero fastest in the direction of the standard basis vector $\mathbf{e}_m$, and we note that $\mathbf{e}_m$ is also the eigenvector of the diagonal B corresponding to the eigenvalue b_m .

2.4.1 Analytic calculation of eigenvectors

In order to test the hypothesis of the EOF method, we calculate the EOF eigenvectors analytically. First, we calculate the empirical mean $\bar{\mathbf{x}}$ of the time series $\{\mathbf{x}_k\}_{k=0}^{N-1}$, defined by equation 2.38, before applying the mean-centring step of the EOF method. We note that the system

equation can be rewritten explicitly as

$$\mathbf{x}_k = B^k \mathbf{x}_0 + \sum_{s=0}^{k-1} B^s S \eta_{k-s}, \quad (2.40)$$

and that this can then be further rewritten to remove the dependence on k from the sum:

$$\mathbf{x}_k = B^k \mathbf{x}_0 + \left[\sum_{s=0}^{\infty} B^s S \eta_{k-s} - \sum_{s=k}^{\infty} B^s S \eta_{k-s} \right] \quad (2.41)$$

$$= B^k \mathbf{x}_0 + \left[\sum_{s=0}^{\infty} B^s S \eta_{k-s} - \sum_{s=0}^{\infty} B^{s+k} S \eta_{-s} \right] \quad (2.42)$$

$$= \sum_{s=0}^{\infty} B^s S \eta_{k-s} - B^k \left(\sum_{s=0}^{\infty} B^s S \eta_{-s} - \mathbf{x}_0 \right) \quad (2.43)$$

There remains a dependence on k inside the first sum in equation 2.43, but this only affects the noise term and is easier to deal with. We also replace k with $N_0 + k$, effectively beginning our analysis of the system after N_0 steps, which will allow us to consider the long-term influence of the initial point $\mathbf{x}_0$. Now to find the empirical mean $\bar{\mathbf{x}}$, that is, we take the sum over k in equation 2.43:

$$\bar{\mathbf{x}} = \frac{1}{N} \sum_{k=0}^{N-1} \mathbf{x}_{N_0+k} = \sum_{s=0}^{\infty} B^s S \frac{1}{N} \sum_{k=0}^{N-1} \eta_{N_0+k-s} - B^{N_0} \frac{1}{N} \sum_{k=0}^{N-1} B^k \left(\sum_{s=0}^{\infty} B^s S \eta_{-s} - \mathbf{x}_0 \right). \quad (2.44)$$

Note that $\frac{1}{n} \sum_{k=0}^{n-1} \eta_{N_0+k-s} \rightarrow \mathbb{E}(\eta_0)$ since it's the empirical mean of independent noise terms, in this case the noise is Gaussian with mean zero so we ignore the first term. Also note that

$$\sum_{k=0}^{N-1} B^k = (I - B)^{-1} (I - B^N), \quad (2.45)$$

where I is the identity matrix, thus we can write

$$\frac{1}{N} \sum_{k=0}^{N-1} \mathbf{x}_{N_0+k} = B^{N_0} \frac{1}{N} (I - B)^{-1} (I - B^N) \left(\mathbf{x}_0 - \sum_{s=0}^{\infty} B^s S \eta_{-s} \right). \quad (2.46)$$

Taking the L^2 -norm of each side and noting that $\|I - B^N\| \leq 1$, the expression becomes

$$\left\| \frac{1}{N} \sum_{k=0}^{N-1} \mathbf{x}_{N_0+k} \right\| \leq \frac{\|B\|^{N_0}}{N} \left(\| (I - B)^{-1} \| \left\| \mathbf{x}_0 - \sum_{s=0}^{\infty} B^s S \eta_{-s} \right\| \right), \quad (2.47)$$

I think here round brackets would be suitable

and we note that the part in square brackets does not depend on N , N_0 or k . We take the expectation of both sides and define

$$K := \mathbb{E} \left[\left\| (I - B)^{-1} \left\| \mathbf{x}_0 - \sum_{s=0}^{\infty} B^s S \eta_{-s} \right\| \right\| \right] \quad (2.48)$$

here square brackets are good, for the expectation

to say that for any value $c > 0$ there is a choice of N and N_0 such that

$$\mathbb{E} [\bar{\mathbf{x}}] \leq K \frac{\|B\|^{N_0}}{N} \leq c. \quad (2.49)$$

With this choice of N and N_0 , we continue to use $\bar{\mathbf{x}} = 0$. In effect, there exists a number N_0 such that after this number of iterations, the influence of the initial point $\mathbf{x}_0$ is no longer visible.

significant

The covariance C , given by

$$C = \mathbb{E} \left[(\mathbf{x}_{k+N_0} - \mathbb{E}(\mathbf{x})) (\mathbf{x}_{k+N_0} - \mathbb{E}(\mathbf{x}))^\top \right], \quad (2.50)$$

is calculated using the empirical mean to approximate the expected value as part of the EOF method by taking the sum over k . We also use the result $\bar{\mathbf{x}} = 0$ to leave

$$C = \frac{1}{N} \sum_{k=0}^{N-1} \left[(\mathbf{x}_{k+N_0}) (\mathbf{x}_{k+N_0})^\top \right]. \quad (2.51)$$

Expanding this using equation 2.43, most of the terms are essentially similar to the expression for $\bar{\mathbf{x}}$ and are similarly approximately zero for the same choice of N and N_0 . This is easily noticed if we simplify the expanded expression using

$$\mathbf{p}_k := \sum_{s=0}^{\infty} B^s S \eta_{k-s}, \quad (2.52)$$

$$\mathbf{q} := \sum_{s=0}^{\infty} B^s S \eta_{-s} - \mathbf{x}_0, \quad (2.53)$$

which allows us to write $\mathbf{x}_k = \mathbf{p}_k - B^k \mathbf{q}$. The terms inside the sum in equation 2.51 are then

$$(\mathbf{p}_{k+N_0}) (\mathbf{p}_{k+N_0})^\top + B^{k+N_0} \mathbf{q} (B^{k+N_0} \mathbf{q})^\top - B^{k+N_0} \mathbf{q} (\mathbf{p}_{k+N_0})^\top - (\mathbf{p}_{k+N_0}) (B^{k+N_0} \mathbf{q})^\top. \quad (2.54)$$

All terms except for the first involve multiplication by the matrix $B^{k+N_0} \mathbf{q}$, or its transpose. Taking the sum over k gives the same expression for the mean $\bar{\mathbf{x}}$ as in equation 2.46, and we have chosen N_0 such that $\bar{\mathbf{x}} = 0$. We now expand the remaining term inside the sum, that is,

expand

$$(\mathbf{p}_{k+N_0})(\mathbf{p}_{k+N_0})^\top = \sum_{s=0}^{\infty} \sum_{r=0}^{\infty} B^s S (\boldsymbol{\eta}_{k+N_0-s})(\boldsymbol{\eta}_{k+N_0-r})^\top S^\top (B^\top)^r. \quad (2.55)$$

The expected value of the matrix $(\boldsymbol{\eta}_{k+N_0-s})(\boldsymbol{\eta}_{k+N_0-r})^\top$ is the zero matrix for $r \neq s$ and the identity matrix for $r = s$. Therefore

$$\mathbb{E}[C] = \mathbb{E} \left[\sum_{k=0}^{N-1} (\mathbf{p}_{k+N_0})(\mathbf{p}_{k+N_0})^\top \right] \quad (2.56)$$

$$= \sum_{k=0}^{N-1} \mathbb{E} \left[(\mathbf{p}_{k+N_0})(\mathbf{p}_{k+N_0})^\top \right] \quad (2.57)$$

$$= \sum_{k=0}^{N-1} \sum_{s=0}^{\infty} B^s S S^\top (B^\top)^s \quad (2.58)$$

$$= \sum_{s=0}^{\infty} B^s S S^\top (B^\top)^s. \quad (2.59)$$

In the special case that $B = \text{diag}(b_1, b_2, \dots)$ and $S = \text{diag}(s_1, s_2, \dots)$, we are able to calculate the entire matrix component-wise to get

$$\mathbb{E}([C]_{ii}) = s_i^2 \frac{1}{N-1} \sum_{r=1}^N \sum_{t=1}^r b_i^{2(t-r)} \quad (2.60)$$

$$= s_i^2 \frac{1}{N-1} \sum_{r=0}^{N-1} (N-r) b_i^{2r} \quad (2.61)$$

$$= s_i^2 \sum_{r=0}^N b_i^{2r} - s_i^2 \sum_{r=0}^N \frac{r-1}{N-1} b_i^{2r} \quad (2.62)$$

$$= s_i^2 \sum_{r=0}^{\infty} b_i^{2r} - s_i^2 \sum_{r=0}^{\infty} \frac{r-1-(r+1)b^{2(N+1)}}{N-1} b_i^{2r}. \quad (2.63)$$

However, when substituting these diagonal matrices into our expression for the covariance matrix in equation 2.59 we get simply

$$\mathbb{E}([C]_{ii}) = s_i^2 \sum_{r=0}^{\infty} b_i^{2r}. \quad (2.64)$$

This is because our working introduced the N_0 term in order to ignore the first terms of the series and approximate the mean as zero. It should be noted that equation 2.59 is therefore only a first approximation to the covariance.

We also note that the same calculation can be done with a time lag so that we are finding the covariance of $\mathbf{x}_k$ and $\mathbf{x}_{k+l}$. In this case the equation 2.55 becomes

$$(\mathbf{p}_{k+N_0})(\mathbf{p}_{k+l+N_0})^\top = \sum_{s=0}^{\infty} \sum_{r=0}^{\infty} B^s S(\eta_{k+N_0-s})(\eta_{k+l+N_0-r})^\top S^\top (B^\top)^r, \quad (2.65)$$

and the expected value of the matrix $(\eta_{k+N_0-s})(\eta_{k+l+N_0-r})^\top$ is equal to the identity matrix when $r = s + l$, rather than $r = s$, and zero otherwise. So we have simply

$$\mathbb{E}(C_{\text{lag-}l}) = \mathbb{E}[(\mathbf{p}_{k+N_0})(\mathbf{p}_{k+l+N_0})^\top] = \sum_{s=0}^{\infty} B^s S S^\top (B^\top)^{s+l}, \quad (2.66)$$

which is the same matrix as the regular (lag-0) covariance given in equation 2.59 but right-multiplied by $(B^l)^\top$. For simplicity, we call this matrix D :

$$D := \mathbb{E}(C) = \sum_{s=0}^{\infty} B^s S S^\top (B^\top)^s. \quad (2.67)$$

This equation allows us to define D intrinsically:

$$D = S S^\top + B D B^\top, \quad (2.68)$$

which we can then solve component-wise. In the fairly trivial one-dimension case $d = s^2 + b^2 d$ we have

$$d = \frac{s^2}{1 - b^2}. \quad (2.69)$$

The two-dimensional case is more interesting and is dealt with more fully in appendix ??, here we restrict our study to the case where B is diagonal, which is equivalent to asking that B is diagonalisable. We obtain

$$D = \begin{bmatrix} \frac{s_{12}^2 + s_{11}^2}{1 - b_{11}^2} & \frac{s_{12}s_{22} + s_{11}s_{21}}{1 - b_{11}b_{22}} \\ \frac{s_{12}s_{22} + s_{11}s_{21}}{1 - b_{11}b_{22}} & \frac{s_{22}^2 + s_{21}^2}{1 - b_{22}^2} \end{bmatrix}, \quad (2.70)$$

for D , and are able to examine the eigenvalues easily because of its symmetry. The eigenvalues of a general symmetric matrix

$$\begin{pmatrix} a & b \\ b & c \end{pmatrix} \quad (2.71)$$

are

$$\frac{1}{2} \left(a + c \pm \sqrt{(a - c)^2 + 4b^2} \right) \quad (2.72)$$

and we can see that, when matrix D is substituted into this, the positive square root will give the largest (principle) eigenvalue. Note that for $b = 0$ (i.e. the matrix is diagonal) the difference between the two eigenvalues is $|a - c|$, but for non-zero b this difference increases, i.e. the eigenvalues become more distinct. Matrix D is diagonal when two entries of S are zero, not in the same column, in which case the system (equation 2.38) is simply two separate AR(1) models. The eigenvector corresponding to the largest eigenvalue (taking the positive root) is

$$\begin{bmatrix} \frac{s_{12}^2 + s_{11}^2}{1 - b_{11}^2} - \frac{s_{22}^2 + s_{21}^2}{1 - b_{22}^2} + \sqrt{\left(\frac{s_{12}^2 + s_{11}^2}{1 - b_{11}^2} - \frac{s_{22}^2 + s_{21}^2}{1 - b_{22}^2} \right)^2 + 4 \left(\frac{s_{12}s_{22} + s_{11}s_{21}}{1 - b_{11}b_{22}} \right)^2} \\ 2 \left(\frac{s_{12}s_{22} + s_{11}s_{21}}{1 - b_{11}b_{22}} \right) \end{bmatrix} \quad (2.73)$$

This is the principle eigenvector used in the EOF method; the assumption is that projecting onto this eigenvector, which maximises variance, captures the interesting behaviour of the system. For the purpose of tipping point analysis, we would therefore expect that this eigenvector has something to do with the eigenvector corresponding to the largest eigenvalue of B (closest to one), since this is the direction in which the system travels slowest to zero. A bifurcation occurs when one eigenvalue approaches one.

Without loss of generality, we assume that the first eigenvalue of B is approaching one; since B is diagonal this is equal to the first entry of B , b_{11} , and the corresponding eigenvector is the standard basis vector $\mathbf{e}_1 = [1, 0]^T$. We make the choice that $b_{11} \approx 1$ (from below), then $1 - b_{11}$ is very small. In appendix ?? we expand the terms in the principle eigenvector (equation 2.73) in leading order terms of $(1 - b_{11})$. If we ignore terms in $\mathcal{O}(1 - b_{11})$ then the principal eigenvector becomes

$$\begin{bmatrix} \frac{s_{12}^2 + s_{11}^2}{1 - b_{11}^2} - \frac{s_{22}^2 + s_{21}^2}{1 - b_{22}^2}, \frac{s_{12}s_{22} + s_{11}s_{21}}{1 - b_{22}} \end{bmatrix}^T \quad (2.74)$$

which, since $1/(1 - b_{11}^2)$ is very large, is approximately equivalent to $[1, 0]^T$ as expected. If S is diagonal ($s_{12} = s_{21} = 0$) then the vector ~~actual~~ is equivalent to $[1, 0]^T$. However, if s_{11}, s_{12} are very small and s_{21}, s_{22} are very large, this may not be true. This implies that there is very large variability in the second component of the system (in the direction of $[0, 1]^T$), and so this exception is predicted by our knowledge of the EOF method.

✓ We have considered the expected value of the covariance matrix, since the noise will cause every realisation of the system to be different. This analysis is also intended to tell us something *general ?*
 ✓ ~~generally~~ *analysis* about the validity of using dimension reduction by the EOF method prior to using tipping point ~~detection or prediction~~ techniques. The method must be valid where the entries of S are large, giving large variability due to noise, or in situation when given time series data of unknown origin which resembles the output of the general system in equation 2.38. In these situations the covariance matrix C obtained from the data by the EOF method may be significantly different to the expectation $D = \mathbb{E}(C)$, in which case it may be uncertain which is truly the principal eigenvalue, especially when the eigenvalues are very similar. Equation 2.72 tells us that the difference between the eigenvalues is smallest when the covariance matrix is diagonal, i.e. when the system is reducible to separate AR(1) models, and that the difference increases as the off-diagonal entries of S increase, i.e. as the system becomes more strongly coupled.

2.4.2 Numerical verification

✓ The analysis in section 2.4.1 makes some assumptions to aid the calculation. In particular, only two-dimensional systems are considered when calculating the eigenvectors of the covariance matrix. In this section we provide numerical experiments to further validate the conclusions of section 2.4.1 and also to provide some insight into three- and four-dimensional systems, ~~although analytic justification has not been provided.~~

Two-dimensional system

We first consider the two-dimensional system

$$\mathbf{x}_k = \Lambda \mathbf{x}_{k-1} + S \eta_k, \quad (2.75)$$

where η_k is a white noise vector and $\Lambda = \text{diag}(\lambda_1, \lambda_2)$ is diagonal. $\lambda_2 = 0.9$ and λ_1 is made to approach one from below:

$$\lambda_1(k) = 0.9 + 10^{-6}k, \quad (2.76)$$

✓ so that $\lambda_1 = \lambda_2$ at $k = 0$ and at $k = 10^5$ a bifurcation occurs as the system tips from convergence to divergence, diverging to infinity in the x_1 direction, i.e. in the direction of $(\pm 1, 0)^\top$. In the previous section we concluded that the nature of the matrix S , specifically the values on the off-diagonal, determines the effectiveness of the EOF method in this context. As an illustration, we initially consider two extreme cases:

1. The two variables are entirely uncoupled: $S_1 = \frac{1}{20} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$.

2. The two variables are coupled: $S_2 = \frac{1}{20} \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$.

(*) may need a number as equation?

The factor $1/\sqrt{2}$ in S_2 is to ensure that the standard deviation of the noise is the same in both cases. Naïvely, we may expect that the leading EOF eigenvector will be similar to $(\pm 1, 0)^\top$ since this is the direction in which the divergence happens, or, before the bifurcation, the direction in which the convergence is weakest and therefore the direction with the longest recovery time after a perturbation from the equilibrium. This may be true eventually but not when λ_1 and λ_2 are very close. According to equation 2.73, the eigenvector, when using the identity matrix S_1 , will be

$$\begin{bmatrix} \frac{1}{1-\lambda_1^2} - \frac{1}{1-0.9^2} \\ 0 \end{bmatrix}, \quad (2.77)$$

which is in the direction $(\pm 1, 0)^\top$ for $\lambda_1 \neq 0.9$, but is the zero vector when $\lambda_1 = 0.9$. At this point, the EOF eigenvector is determined only by the noise, and therefore the argument angle of that vector will be random in the range $[-\pi/2, \pi/2]$, with mean zero. The angle will tend to zero (i.e. the EOF eigenvector will approach $(1, 0)^\top$) as the difference $|\lambda_1 - 0.9|$ is sufficiently large that the EOF method is able to distinguish the effect upon the system from the noise.

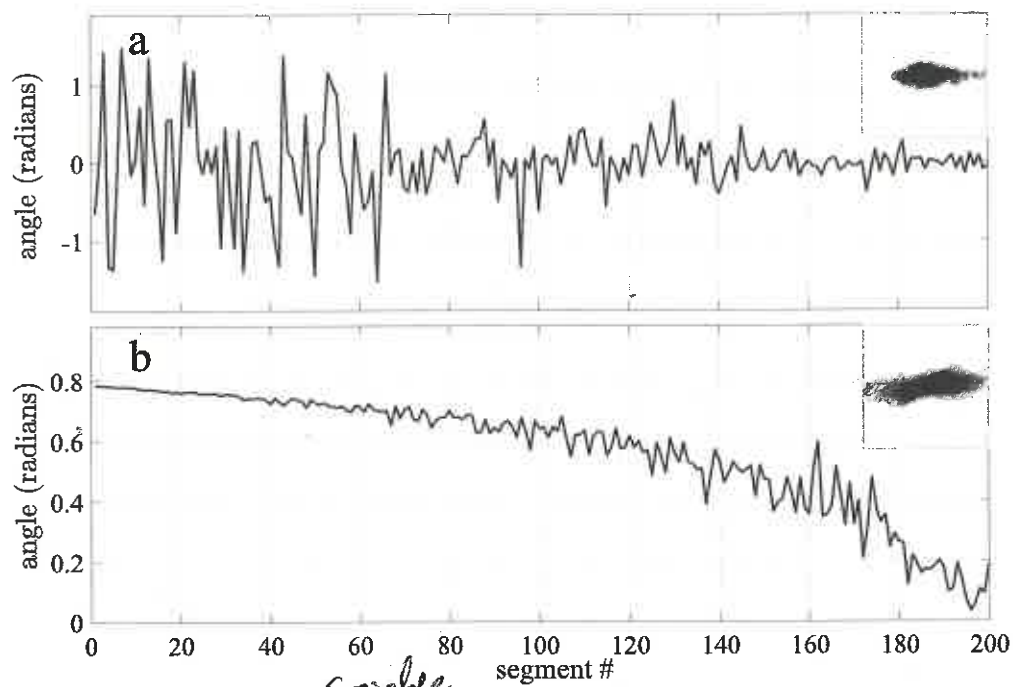
When using the matrix S_2 we will have the vector

$$\begin{bmatrix} \frac{1}{1-\lambda_1^2} - \frac{1}{0.19} + \sqrt{\left(\frac{1}{1-\lambda_1^2} - \frac{1}{0.19}\right)^2 + \left(\frac{2}{1-0.9\lambda_1}\right)^2} \\ \frac{2}{1-0.9\lambda_1} \end{bmatrix} \quad (2.78)$$

which is in the direction $(1, 1)^\top$ for $\lambda_1 = 0.9$ and approaches $(1, 0)^\top$ as $\lambda_1 \rightarrow 1$ as the $1/(1-\lambda_1^2)$ term becomes very large. The argument angle of this vector will therefore start as $\theta_{\text{eig}} = \pi/4$ when $k = 0$ (i.e. $\lambda_1 = 0.9$) and tends towards zero as k increases.

Using both of these matrices S_1 and S_2 we calculate the terms $\mathbf{x}_k$ up to the point of the bifurcation ($k = 10^5$). For each of the resulting series we perform the EOF analysis on 200 half-overlapping segments of length 10^3 , in order to see how the EOF eigenvectors change as k increases. In each segment the argument angle of the EOF eigenvector is calculated.

Figure 2.8 shows the result when using S_1 (panel a) and S_2 (panel b). We see that the nature of the matrix S does affect the EOF projection. When the two systems are uncoupled (i.e. the



✓ Fig. 2.8 The ~~argument~~ ^{variable} angle of the EOF eigenvector calculated in 200 overlapping segments of length 10^3 . Two cases of the system in equation 2.75 are considered: using S_1 (panel a) and using S_2 (panel b). The series itself is plotted, in each case, in the insert at the top-right of the panel, both x and y axes range from -2 to 2 .

see (*)

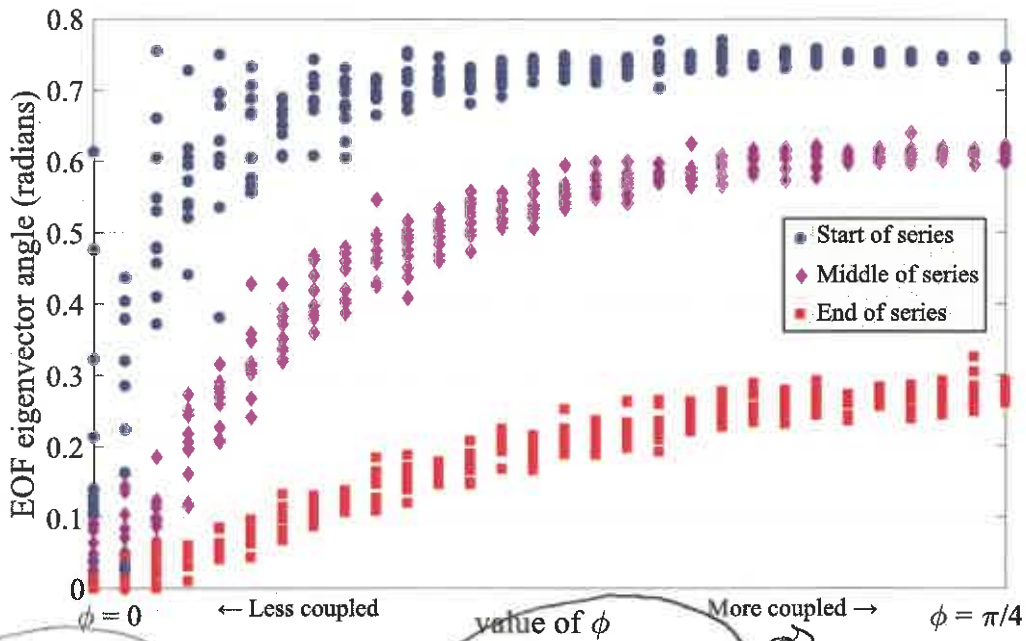


Fig. 2.9 Using a range of matrices $S^{(\phi)}$ the absolute value of the argument angle of the EOF eigenvector calculated in three segments of the series $\mathbf{x}_k$ (start, middle and end). The series is calculated 10 times for each matrix $S^{(\phi)}$, showing the range of angles, particularly for low values of ϕ where the system variables are uncoupled.

two variables are two separate AR(1) models) the angle of the eigenvector is apparently random at first when λ_1 is close to the value 0.9 but the angle converges towards zero as λ_1 goes to 1, as predicted. When the two variables are coupled, the argument angle of the EOF eigenvector starts at $\pi/2$ (for $\lambda_1 = 0.9$) and tends towards zero, also as predicted.

We now perform the same analysis using a range of 30 intermediate symmetric matrices

$$S^{(\phi)} = \frac{1}{20} \begin{pmatrix} \cos(\phi) & \sin(\phi) \\ \sin(\phi) & \cos(\phi) \end{pmatrix}, \quad (2.79)$$

where ϕ varies from zero (giving the identity matrix S_1) to $\pi/4$ (giving the ones matrix S_2). For each of the 30 matrices $S^{(\phi)}$ the series $\mathbf{x}_k$ is calculated and the EOF eigenvector argument angle is found in three segments: one at the start of the series where λ_1 is close to 0.9 ($k \in [1 \times 10^4, 2 \times 10^4]$); one in the middle of the series ($k \in [4.5 \times 10^4, 5.5 \times 10^4]$); and one close to the bifurcation ($k \in [8 \times 10^4, 9 \times 10^4]$). This is repeated ten times to overcome the effect of noise, and only the absolute value of the argument angle is recorded. The result is shown in figure 2.9. We see that the results at $\phi = 0$ (matrix S_1) and $\phi = \pi/4$ (matrix S_2) are

similar to the results shown in figure 2.8, bearing in mind that, this time, the absolute angle is recorded. However, in the case of the uncoupled system (small ϕ) there is less variability here at the start of the series than seen in figure 2.8, rather than being distributed in the range $[0, \pi/2]$ the 'start of series' points, which correspond to the same values of k as segments numbered 20 to 40 in figure 2.8, do not exceed 0.8. This is because of the larger sample used: 10^4 points rather than 10^3 points. Repeating the figure 2.8 analysis with a larger segment size, we note that ~~a larger segment size~~ results in a more accurate EOF projection since, with a larger sample, the effect of the system eigenvalues of the convergence or divergence is more easily distinguishable from the noise.

We note that after about half way along the horizontal axis ($\phi > \pi/8$) there is almost no change in any of the segments. Most of the qualitative change occurs over the first six values of ϕ used, that is $\phi < \pi/20$, which leads us to consider the matrix

$$S^{(\pi/20)} = \frac{1}{20} \begin{pmatrix} 0.988 & 0.156 \\ 0.156 & 0.988 \end{pmatrix}. \quad (2.80)$$

At this level of noise (standard deviation = $1/20$) the degree to which the system variables are coupled does not qualitatively affect the EOF method beyond this point.

Three-dimensional system

We now perform the same analysis as above with equation 2.75 being a three dimensional system, that is, $\Lambda = \text{diag}(\lambda_1, \lambda_2, \lambda_2)$ is diagonal. $\lambda_2 = 0.9$ and λ_1 is made to approach one from below:

$$\lambda_1(k) = 0.9 + 10^{-6}k. \quad (2.81)$$

We use the matrices $S_1 = \sigma I$ and $S_2 = \sigma(1/\sqrt{3})J$ where I is the identity matrix, J is the ones matrix and $\sigma = 1/20$ is the standard deviation. Again, the series $\mathbf{x}_k$ is calculated up to $k = 10^5$ and the EOF eigenvector is found in overlapping sections of length 1000. This time the angle recorded is necessarily the absolute angle between the EOF eigenvector and $(1, 0, 0)^T$, since a negative angle does not make sense in three dimensional space. The result is shown in figure 2.10. We note the obvious similarity to figure 2.8, bearing in mind (looking at panel a) that the absolute value of the angle is now considered, so that the random angle will be distributed in the range $[0, \pi/2]$, and (looking at panel b) that the angle made by the vector $(1, 1, 1)^T$ is $\arccos(1/\sqrt{3}) = 0.9553\dots$ which is the angle of the EOF eigenvector when $\lambda_1 = 0.9$. It appears that there is greater variability in the S_2 case close to the end of the series, but the general pattern is the same.

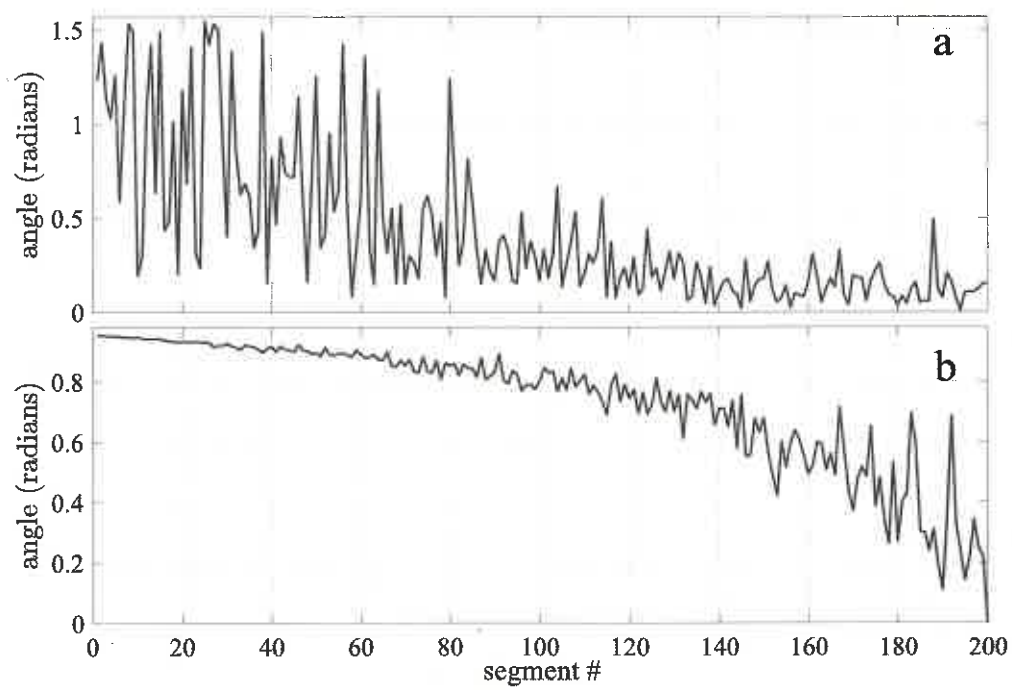
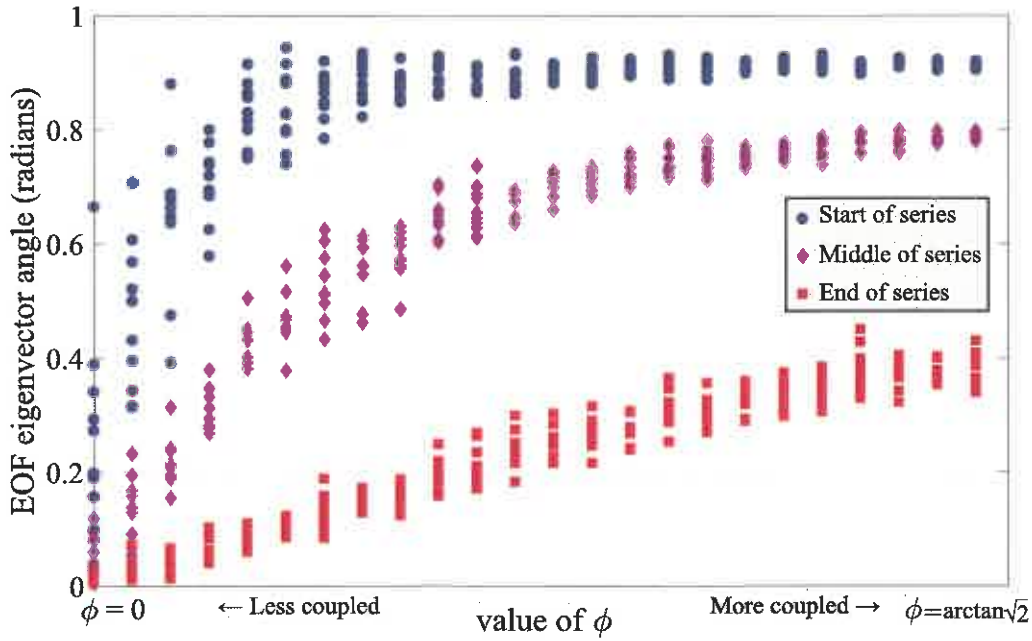


Fig. 2.10 The argument angle of the EOF eigenvector calculated in 200 overlapping segments of length 10^3 . Two cases of the three-dimensional system in equation 2.75 are considered: using S_1 (panel a) and using S_2 (panel b).



caption
to be
equivalent
to Fig
2.9

Fig. 2.11 Using a range of 3×3 matrices $S^{(\phi)}$, the absolute value of the argument angle of the EOF eigenvector calculated in three segments of the series $\mathbf{x}_k$: start, middle and end. The series is calculated 10 times for each matrix $S^{(\phi)}$, showing the range of angles.

As a further confirmation we also repeat the analysis behind figure 2.9 using a three-dimensional system. We use the general 3×3 matrix $S^{(\phi)}$ which has $\cos(\phi)$ on the diagonal and $\sin(\phi)/\sqrt{2}$ everywhere else:

$$S^{(\phi)} = \frac{1}{20} \begin{pmatrix} \cos(\phi) & \frac{1}{\sqrt{2}} \sin(\phi) & \frac{1}{\sqrt{2}} \sin(\phi) \\ \frac{1}{\sqrt{2}} \sin(\phi) & \cos(\phi) & \frac{1}{\sqrt{2}} \sin(\phi) \\ \frac{1}{\sqrt{2}} \sin(\phi) & \frac{1}{\sqrt{2}} \sin(\phi) & \cos(\phi) \end{pmatrix}. \quad (2.82)$$

Similarly to the two-dimensional case, as ϕ increases from zero to $\arctan(\sqrt{2})$ the matrix $S^{(\phi)}$ changes from S_1 , which gives an uncoupled system of separate AR(1) models, to S_2 , which gives a system with an identical noise term in all three variables. The same steps to produce figure 2.9 are followed in the three-dimensional case and the result is shown in figure 2.11. We note, as with the two-dimensional case, that the EOF method is more accurate at the start of the more uncoupled system series in this figure, where the segment sampled is 10^4 points, than in figure 2.10 where the segment size is 10^3 points. We also note again that the EOF eigenvector at the start of the coupled system series is in the direction $(1, 1, 1)^T$ with an angle

$\arccos(1/\sqrt{3}) = 0.9553$ rather than $\arccos(1/\sqrt{2}) = \pi/4$ as in the two-dimensional case (from the direction $(1, 1)^\top$). Again, there is very little change after about half way along the horizontal axis ($\phi > \arctan(\sqrt{2})/2$) and most of the qualitative change occurs over the first five or six values of ϕ used here ($\phi < 0.2$).

Higher dimensions

✓ The results of this same analysis applied to 4 or 5 dimensional systems are similar to the results here and are presented in appendix(??).

DO APPENDICES

2.5 An alternative EOF method for dimension reduction

✓ Although the EOF method is widely used in ecology and geosciences [Hasselmann, 1988; Dommenget and Latif, 2002; Bathiany et al., 2013] occasionally, even, in the process of detecting EWSs [Held and Kleinen, 2004], it is not ideally suited for this purpose because it does not take into account the internal dynamics of the system. Dommenget and Latif [2002] notes.

... EOF analyses have problems in identifying the dominant centers of action or the teleconnections between these centers of action in multivariate datasets. We therefore have to be very careful in interpreting the EOF modes as potential physical modes.

Hasselmann [1988] proposes that the use of principal interaction patterns and principal oscillation patterns (PIPs and POPs) [Von Storch et al., 1995; Kwasniok, 1996] is much better suited to applications where the evolution of a system over time is relevant, including EWS applications. The PIP technique is introduced by Hasselmann [1988] as an alternative to EOF analysis for reducing the dimension of a system to “a few dominant patterns”, with the benefit that the system dynamics are modelled, similarly to an ARMA parametrisation. The use of PIPs and POPs may therefore be used to reduce the dimension of a system prior to applying one-dimensional tipping point indicators, as an alternative to EOFs.

Besides these shortcomings, which lead one to favour PIPs and POPs as an alternative, we have noted (section 2.4) that it may not necessarily be the case that the EOF modes of a system will capture the largest changes in variance, which would be relevant to a variance-based EWS indicator. They are constructed so as to maximise the value of the variance. It is also not necessarily the case that the EOF modes will be relevant to an autocorrelation-based indicator. In this section we present an alternative to the EOF method where, rather than projecting the

✓ ^{together} system ^{on} to a set of vectors such that the variance of the first mode is maximal (as in the typical EOF method), we do so such that the lag-1 autocorrelation is maximal. We investigate the result of using this alternative in the case of the system experiencing a homoclinic bifurcation, described in section 2.3 (see equation 2.23). ✓

An autocorrelation-based EOF method

First, we consider the typical EOF method, this involves taking multiple time series and projecting onto the vector that maximises the variance. As an illustration, consider the two time series $[x_i]_{i=1}^N$ and $[y_i]_{i=1}^N$, which we assume to be mean-centred, i.e.

$$\frac{1}{N} \sum_i x_i = \frac{1}{N} \sum_i y_i = 0 \quad (2.83)$$

and non-dimensionalised. We then construct the matrix

$$X = \begin{pmatrix} x_1 & y_1 \\ x_2 & y_2 \\ \vdots & \vdots \\ x_N & y_N \end{pmatrix}, \quad (2.84)$$

✓ and note that, in general, we may have any number of time series and, correspondingly, any number of columns of X . We now find the unit vector $\underline{u}$, and ~~therefore~~ the time series $\mathbf{p} = [p_i]_{i=1}^N$ which is the projection of the matrix X onto this vector,

$$\mathbf{p} = X\underline{u}, \quad (2.85)$$

such that the variance of $\mathbf{p}$ is maximal given all possible $\underline{u}$.

$$\text{Var}(\mathbf{p}) = \frac{1}{N} \mathbf{p}^T \mathbf{p} = \frac{1}{N} (X\underline{u})^T (X\underline{u}) = \underline{u}^T C_0 \underline{u}, \quad (2.86)$$

where C_0 is the lag-zero auto-covariance matrix of X ,

$$C_0 = \frac{1}{N} \sum_{i=1}^N \mathbf{x}_i^T \mathbf{x}_i = \frac{1}{N} X^T X. \quad (2.87)$$

To maximise the variance, with the constraint $\|\underline{u}\| = 1$, we use Lagrange multipliers with the Lagrange function

$$\mathcal{L}(\underline{u}, \lambda) = \underline{u}^T C_0 \underline{u} - \lambda \left[\left(\sum_i u_i^2 \right) - 1 \right]$$

Calculating the gradient gives

$$\nabla \mathcal{L} = 0$$

$$\frac{\partial}{\partial u_k} (\underline{u}^T C_0 \underline{u}) - \lambda \frac{\partial}{\partial u_k} (\sum u_k^2) = 0$$

$$2C_0 \underline{u} - 2\lambda \underline{u} = 0$$

(2.89)

(2.90)

(2.91)

since C_0 is symmetric, which implies $\underline{u}$ is an eigenvector of C_0 . Say the corresponding eigenvalue is λ_u then

$$\text{Var}(\mathbf{p}) = \underline{u}^T C_0 \underline{u} = \underline{u}^T \lambda_u \underline{u} = \lambda_u \quad (2.92)$$

because $\underline{u}^T \underline{u} = 1$. Thus to maximise $\text{Var}(\mathbf{p})$, we must choose $\underline{u}$ such that λ_u is largest, i.e. $\underline{u}$ is the eigenvector corresponding to the largest eigenvalue of C_0 .

Now, we may also calculate the lag-1 auto-covariance

$$C_1 = \frac{1}{N-1} \sum_{i=2}^N \mathbf{x}_i^T \mathbf{x}_{i-1} = \frac{1}{N-1} X^T S X, \quad (2.93)$$

where S is the matrix such that

$$SX = \begin{pmatrix} 0 \\ \mathbf{x}_1 \\ \mathbf{x}_2 \\ \vdots \\ \mathbf{x}_{N-1} \end{pmatrix}, \quad (2.94)$$

and we have the autocorrelation function

$$\text{ACF}_1(\mathbf{p}) = \frac{\frac{1}{N-1} \sum_i p_i p_{i-1}}{\text{Var}(\mathbf{p})} \quad (2.95)$$

$$= \left(\frac{1}{N-1} (X\underline{u})^T (SX\underline{u}) \right) \left(\frac{1}{N} (X\underline{u})^T (X\underline{u}) \right)^{-1} \quad (2.96)$$

$$= (\underline{u}^T C_1 \underline{u}) (\underline{u}^T C_0 \underline{u})^{-1}. \quad (2.97)$$

if you use separate \begin{equation} instead of array instead, or use \nonumber

Now, instead of choosing a one-dimensional projection $\mathbf{p} = X\mathbf{u}$ such that $\text{Var}(\mathbf{p})$ is maximal, we may want to find the projection such that $\text{ACF}_1(\mathbf{p})$ is maximal. That is, we wish to find $\mathbf{u}_{\max}$ such that

$$\text{ACF}_1(X\mathbf{u}_{\max}) = \max_{\|\mathbf{u}\|=1} [\text{ACF}_1(X\mathbf{u})] \quad (2.98)$$

$$= \max_{\|\mathbf{u}\|=1} \left[\left(\mathbf{u}^\top C_1 \mathbf{u} \right) \left(\mathbf{u}^\top C_0 \mathbf{u} \right)^{-1} \right] \quad (2.99)$$

Attempting to maximise this expression in $\mathbf{u}$ using Lagrange multipliers, as for the simpler EOF problem (equation 2.91), we obtain

$$\nabla \mathcal{L} = 0 \quad (2.100)$$

$$\frac{\partial}{\partial u_k} \left(\frac{\mathbf{u}^\top C_1 \mathbf{u}}{\mathbf{u}^\top C_0 \mathbf{u}} \right) = \lambda \frac{\partial}{\partial u_k} (\sum u_k^2) \quad (2.101)$$

$$\left(\mathbf{u}^\top C_0 \mathbf{u} \right) \frac{\partial}{\partial u_k} \left(\mathbf{u}^\top C_1 \mathbf{u} \right) - \left(\mathbf{u}^\top C_1 \mathbf{u} \right) \frac{\partial}{\partial u_k} \left(\mathbf{u}^\top C_0 \mathbf{u} \right) = 2\lambda \left(\mathbf{u}^\top C_0 \mathbf{u} \right)^2 \mathbf{u} \quad (2.102)$$

$$\left(\mathbf{u}^\top C_0 \mathbf{u} \right) \left(C_1 + C_1^\top \right) \mathbf{u} - 2 \left(\mathbf{u}^\top C_1 \mathbf{u} \right) C_0 \mathbf{u} = 2\lambda \left(\mathbf{u}^\top C_0 \mathbf{u} \right)^2 \mathbf{u} \quad (2.103)$$

which will not result in such an elegant solution and will require a tedious element-wise calculation. However, the maximisation problem can be solved using a computer, either by numerical solution of equation 2.103 or by other numerical optimisation methods applied to equation 2.98, once C_0 and C_1 have been calculated. A simple method to obtain a numerical solution, for problems with a low number of dimensions, is to project the time series onto many unit vectors $\mathbf{u}$ in a systematic way and select the $\mathbf{u}_{\max}$ where the autocorrelation of $\mathbf{p}$ is largest.

Application to bifurcating dynamical systems

The method described ^{above} here is applied to the system experiencing a homoclinic bifurcation, described in section 2.3. The system is given by equations

$$\begin{aligned} \dot{x} &= y + \varepsilon^{(x)}, \\ \dot{y} &= \alpha - x^2 + \varepsilon^{(y)}, \end{aligned} \quad (2.104)$$

where $\varepsilon^{(x)}$, $\varepsilon^{(y)}$ are white noise terms. The system is integrated and sampled with a time step of $\Delta t = 0.5$ to produce a time series of 200 points (in each variable). The ACF-based EOF method is then applied to the two variables to reduce the dimension, yielding a one dimensional time series. To do this, the maximisation problem in equation 2.98 is solved element-wise using Mat-

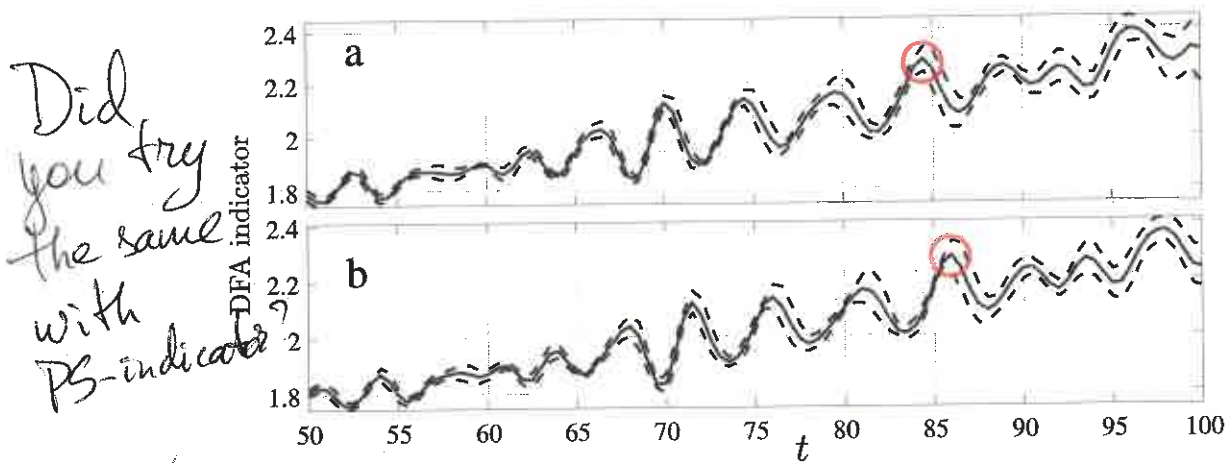


Fig. 2.12 Showing the DFA indicator with a window of 100 points calculated for the one-dimensional time series obtained using the variance-based EOF technique (panel a) and the autocorrelation-based EOF technique (panel b) applied to the system described by equation 2.104, which experiences a homoclinic bifurcation at $t = 100$. In each panel the plot shows the mean over ten realisations of the dynamical system, with error bounds of one standard deviation. One example of a common feature is circled in red.

Matlab
 ✓ Lab's standard `fmincon` interior-point optimisation algorithm [Dennis Jr and Schnabel, 1996; Byrd et al., 1999, 2000] to find $\underline{u}_{\max}$, and the one-dimension projection $X\underline{u}_{\max}$ is calculated, where X is the mean-centred data matrix.

Having obtained the one-dimensional time series $X\underline{u}_{\max}$ via this autocorrelation-based EOF technique, the DFA indicator is calculated with a window size of 100 points: figure 2.12b shows the result; for comparison, the same process is also performed using the regular variance-based EOF technique (panel a). We see that there is little difference in the DFA indicator whichever EOF technique is used, the same features appear in both signals. However, the features appear earlier when the regular, variance-based EOF technique is used (panel a). An example of a shared feature (a peak in the oscillating DFA indicator signal) is circled in red in figure 2.12: this feature occurs before $t = 85$ in panel a but slightly after $t = 85$ in panel b. This is not an artefact of the alternative EOF technique producing a lagged time series due to the lag in the autocorrelation, since only the lag-1 ACF is used and a single time step is of length $\Delta t = 0.5$, whereas the apparent shift in the features of the DFA indicator signal is ≈ 1.5 . Rather, this is simply due to the different projection used to obtain the one-dimensional time series to which the DFA indicator is applied.

To investigate the alternative EOF technique further, we also apply it to the other two-variable dynamical systems described in section 2.3: the system experiencing a Hopf bifurcation

(equation 2.26), to which we refer as "system B"; and the Van der Pol oscillator (equation 2.35), or "system C". The system described in equation 2.104 is referred to as "system A". Each of the three systems is integrated 100 times and both EOF techniques are applied to each resulting two-variable time series. In each case we record the eigenvector onto which the system is projected to obtain the one-dimensional time series. A multivariate time series has, for this purpose, two vectors v_V, v_A , characterised by their argument angles $\theta_V, \theta_A \in (-\pi, \pi]$, associated with it, and onto which we may choose to project it. v_V is the unit vector which maximises the variance, as the vector u in equation 2.90, used in the regular variance-based EOF technique. v_A is the unit vector which maximises the autocorrelation, used in the alternative autocorrelation-based EOF technique. These vectors, for the three dynamical systems, are represented graphically on the half-circle in figure 2.13. We see that in all three systems there is little variability in the variance-maximising vector (displayed in blue), the autocorrelation-maximising vector is more sensitive to differences between different realisations of the system. This effect is visible, in particular, in system A, which experiences a homoclinic bifurcation and is the subject of the study into the DFA indicator (figure 2.12). For each realisation of each system the difference $|\theta_V - \theta_A|$ is calculated and the mean is found to be 0.510 (29.2°) in system A; 0.184 (10.6°) in system B; and 0.162 (9.30°) in system C. That is, in all three systems, the vector v_A is closer to v_V than the perpendicular; particularly in systems B and C there will be little observable difference between the two projected time series.

2.6 EWS in spatially separated data

I AM CONSIDERING PLACING THIS SECTION IN A SEPARATE CHAPTER ALONG WITH THE FOLLOW-UP WORK ABOUT THE TROPICAL CYCLONE, SINCE THIS IS SOMEHOW DIFFERENT TO THE REST OF THIS CHAPTER.

Besides techniques which allow one to detect or predict a tipping point in a multivariate system, we are also interested in dynamical systems evaluated over a field. In a scenario where one has, for example, many time series associated with locations over a 2D field, one may reduce the dimension of the system using the EOF method to condense all of the available information. However, one may also wish to use the spatial locations of the data, for example, to predict the location of the disturbance which is causing the tipping point.

In this section we present a method to visualise the behaviour of one-dimensional tipping point indicators evaluated at multiple locations, at a specific time slice, thus allowing one to observe locations, or hotspots, where a tipping point is likely to occur.

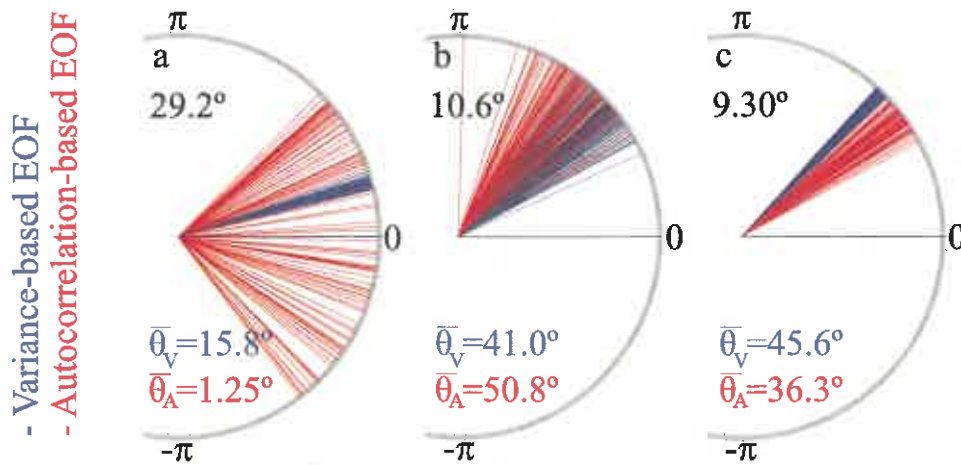


Fig. 2.13 Showing the eigenvectors on the unit half-circle used to obtain the projection involved in the EOF technique (blue) and the alternative autocorrelation-based EOF technique (red) applied to 100 realisations of each dynamical system. Panel a: system A (equation 2.104). Panel b: system B (equation 2.26). Panel c: system C (equation 2.35). The mean difference (in degrees) in the argument angles, $|\bar{\theta}_V - \bar{\theta}_A|$ is displayed in black font beneath the identifying panel letter. The mean angles $\bar{\theta}_V$ and $\bar{\theta}_A$ are displayed in their respective colours.

A 2D visualisation of indicator behaviour

In figure 2.14 we illustrate an example of a propagating front over a 2D field. The time series $z_{xy}(t)$ is evaluated at each point on the field, and a tipping point occurs at the front (double line) which propagates from top-right to bottom-left. If we take the time series at points **A** and **B** and perform an EWS analysis, for example calculating the ACF1 indicator in a sliding window, we expect that the EWS will be stronger at point **B** than at point **A** because the tipping is closer (in time) to **B**. Given some knowledge of the nature of the system one could conclude that the front is approaching from the right (from the positive x direction), and performing the analysis for many more points in a grid over the field, one could determine more precisely the shape of the front and its direction.

It will be difficult to visualise an ACF1 indicator series at every point on the grid (assuming the grid is larger than 3-by-3) in order to determine at which points the EWS is strongest. It is therefore useful to measure the increase in the indicator value at a specified point in time and to create a contour plot of this measure over the grid.

As an example, we consider a system with a pitchfork bifurcation, which was used as an example in section 1.4. In this example, many similar dynamical systems are defined over the

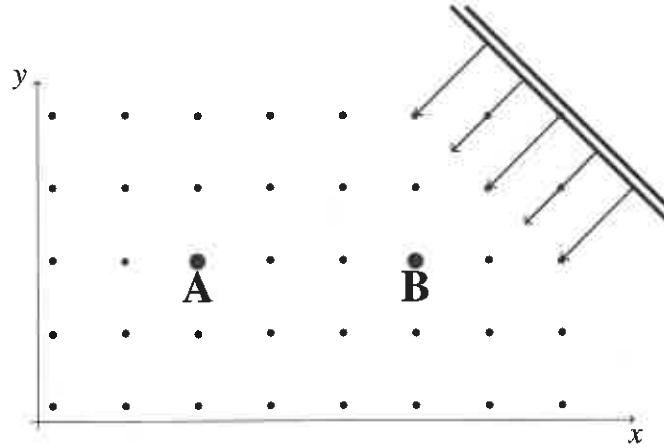


Fig. 2.14 Propagating front. The time series $z_{xy}(t)$ is evaluated at each point on the field, and a tipping point occurs at the front (double line) which propagates from top-right to bottom-left. We expect that an EWS of the tipping point will be stronger at point **B** than at point **A** because the tipping is closer (in time) to **B**.

discrete xy plane $\mathbb{Z}^2$, by the equation

$$\frac{dz_{xy}}{dt} = -\frac{\partial}{\partial z_{xy}} \left(z_{xy}^4 + \frac{\mu - x - y}{10} z_{xy}^2 \right) + \sigma \eta_{xy}, \quad (2.105)$$

where the η_{xy} are independently distributed Gaussian noise series. The line $y = -x + \mu$ defines the front: if $x + y < \mu$ the z^2 coefficient is positive and the system $z_{xy}(t)$ has a double-well potential function; if $x + y \geq \mu$ then the system has a single-well potential.

We create a 10-by-10 grid and integrate equation 2.105 at each point (x, y) from $t = 0$ to $t = 1000$ with time step $\Delta t = 0.05$. μ is made to decrease linearly at each time step so that at $t = 0$, $\mu = 20$ and at $t = 1000$, $\mu = 10$. The series is sampled, after the integration, at intervals of one time unit to give a series of length 1000 points. For each time series z_{xy} , the ACF1 indicator is calculated in a sliding window of 100 points. In order to determine the location of the moving front at time t , one must determine whether the ACF1 indicator is increasing at each point (x, y) . We use the Mann-Kendall coefficient given by the equation

$$\text{coefficient}(X) = \sum_{i=1}^{N-1} \sum_{j=i+1}^N \text{sign}(X_j - X_i), \quad (2.106)$$

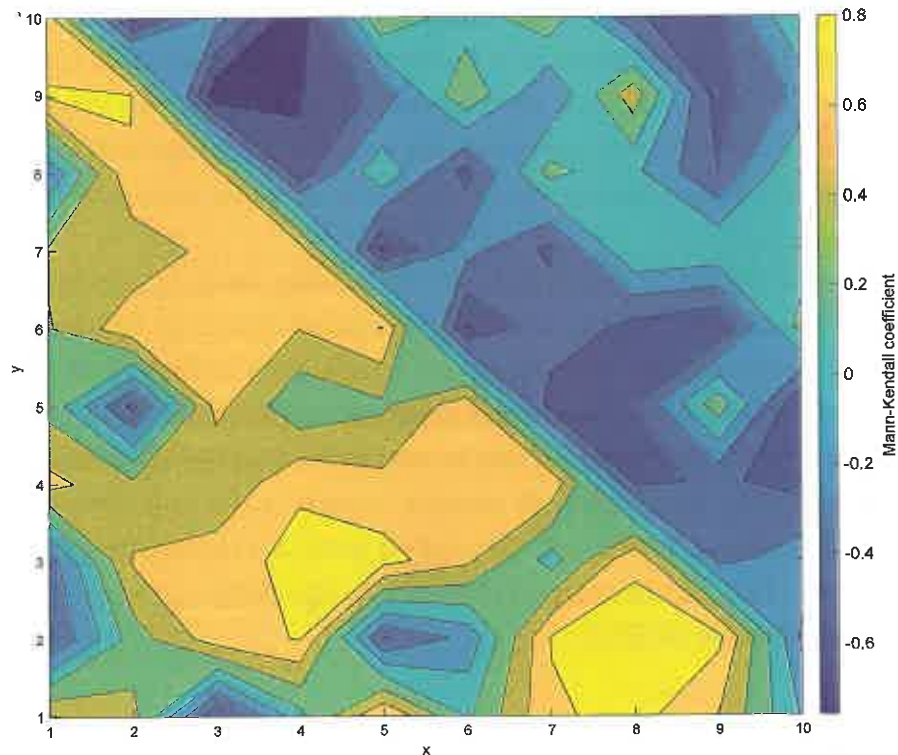
for a time series X . This coefficient is calculate for each indicator series in the range $[t - 100, t]$, thus showing whether the indicator series is increasing (positive coefficient) or decreasing (negative) in the 100-point window prior to time t . Figure 2.15 shows the result when $t = 1000$,

that is, the time when $\mu = 10$ and the pitchfork bifurcation is occurring in systems z_{xy} on the line $x + y = 10$. To the top-right of this line, the bifurcation has already occurred and the lag-1 ACF is decreasing (negative Mann-Kendall coefficient). To the bottom-left of this line, the systems have not yet bifurcated and the ACF1 indicator is increasing (positive Mann-Kendall coefficient) as an EWS of the approaching tipping point. As demonstrated in the simple application of the ACF1 indicator to the pitchfork bifurcating system in section 1.4, there is some variability in the indicator series and we cannot expect a completely smooth gradient to appear in the contour plot. However, the figure 2.15 does accurately represent the existence of a front. We note that a clear line between positive and negative appears to exist along the line $x + y = 11$, because of the lag effect caused by the window size of 100 points used in the calculation of both the ACF1 indicator and the Mann-Kendall coefficient. This effect can be taken into account to conclude that the bifurcation is actually occurring some way in before the front apparent in the figure. The speed at which the front is advancing can be ascertained by taking snapshots of the system at intervals, that is, using a varying values of t when calculating the Mann-Kendall coefficient in the preparation of the contour plot.

If the experiment is repeated several times and the Mann-Kendall coefficient is calculated for the mean over all ACF1 indicator series at each point, the variability in the indicator will be smoothed and the line representing the front will be clearer. This may be necessary for systems where the indicator series contains greater variability, but impractical for applications where the experiment only occurs once.

2.7 Discussion

In this chapter we have presented ~~two~~ techniques for detecting early warning signals in times series of more than one variable. The first technique, introduced by Williamson and Lenton [2015], reconstructs the eigenvalues of the system's Jacobian matrix using the relationship to the autocorrelation matrix. These eigenvalues are then used as indicators of a tipping point based on ~~some basic assumption~~ about the nature of the tipping. For example, a Hopf bifurcation in a two-variable system occurs when the Jacobian eigenvalues cross the imaginary axis, so calculating the behaviour of the imaginary part over time will provide a useful indicator which predicts a bifurcation as it approaches zero. The other technique presented ~~is that of using~~ EOFs to reduce the dimension of the system, before applying a familiar one-dimensional indicator to the resulting one-dimensional time series. We have applied both of these techniques to time series data from three known two-variable dynamical systems. In all cases the DFA indicator applied to the first EOF series provided a clear EWS in the mean. The ACF and PS indicators applied to the first EOF series did not give such a clear signal. The reconstructed Jacobian



✓ ^{2D} Fig. 2.15 Bifurcations caused by an approaching front. At each point (x, y) a system is described by equation 2.105 where the value of μ is decreasing. The figure shows the Mann-Kendall coefficient of the ACF1 indicator of the series when $\mu = 10$, when the bifurcation is occurring at points on the line $y + x = 10$. To the top-right of this line, the bifurcation has already occurred and the lag-1 ACF is decreasing (negative Mann-Kendall coefficient). To the bottom-left of this line, the systems have not yet bifurcated and the ACF1 indicator is increasing (positive Mann-Kendall coefficient) as an EWS of the tipping point.

eigenvalues did provide an EWS in some cases, but it is necessary to know which eigenvalue to use, this is not a problem for a two-variable system.

We noted that the use of EOFs to reduce the dimension of a system, in an EWS context, assumes that the relevant signal will not be lost in the process. The EOF method projects a time series in such a way as to maximise the variance, which may not maximise the increase in the relevant tipping point indicator. We calculated the EOF scores analytically of a general linear dynamical system. We found that, except in contrived cases with very large variance in one variable, the EOF projection vector is similar to the direction in which the system diverges at a tipping point. We also devised an alternative EOF method which maximises the lag-1 autocorrelation of the projection rather than the variance. When applied to three dynamical systems we found that the projection vector was similar to that for the regular EOF method, but there was greater variability.

In addition to this work, which concentrates on time series in several variables, with examples of two-variable systems, we have also studied dynamical systems defined over a discrete two-dimensional field. This type of data is particularly common in meteorology where data is collected at discrete locations over a geographic area (including the entire globe). We have proposed a simple method which may be used to visualise tipping point indicators over a field, and have applied this to a simple example based on a simple pitchfork bifurcation for which a reliable EWS can be obtained. The method works as expected, but this is a simple test: this is a method to aid the visualisation of the indicator, and it is necessary that the indicator used is known to provide an EWS.

Further work

In the section we have repeatedly used three dynamical systems as examples. It is always possible to apply the techniques presented here to more systems with different types of tipping.

This work has been done with an aim of demonstrating the validity of using the various techniques, rather than with a view to studying early warning signals in a comprehensive collection of dynamical systems. Indeed, when meteorological data is presented, the type of tipping is often unknown: that is, it is not known whether the tipping is caused by, for example, a Hopf bifurcation in the underlying system. It is therefore useful to have a variety of techniques available, that could be compared to infer about the possible nature of the critical transition.

When calculating the EOF eigenvectors analytically for the general linear system, we did not ~~also~~ calculate an error term by considering the variance in the system, we have only considered the mean. This does not appear to be possible without making further simplifications to the equations, but it may be possible to provide an estimate after further working.

in further work.

✓ ~~As with the work concerning multi-variable time series, it is always possible to include more examples for the work concerning systems on a two-dimensional field.~~ It would be particularly interesting to use a general, simple system which has similarities to a meteorological event. In the next chapter, a specific model of a tropical cyclone is presented as a further test of the technique.

is relevant